

Final Project

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Introduction

In financial services, accurately predicting loan approvals is essential for mitigating risk and ensuring equitable lending practices. This project aims to develop a predictive model for loan approval decisions, utilizing a dataset containing various applicant attributes.

The initial phase involved comprehensive data preprocessing: handling missing values, encoding categorical variables, and normalizing numerical features. Exploratory data analysis was conducted to discern patterns and relationships within the data, including the visualization.

Subsequently, dimensionality reduction techniques, such as Principal Component Analysis (PCA), were applied to distill the most informative features, thereby enhancing model performance and interpretability. The modeling phase encompassed both supervised and unsupervised learning approaches. Supervised methods included logistic regression, decision trees, and k-nearest neighbors (KNN), while unsupervised techniques involved clustering algorithms like k-prototypes and hierarchical clustering based on Gower distance.

Import Data

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## 'data.frame':   598 obs. of  13 variables:
## $ Loan_ID       : chr  "LP001002" "LP001003" "LP001005"
## "LP001006" ...
## $ Gender        : chr  "Male" "Male" "Male" "Male" ...
## $ Married       : chr  "No" "Yes" "Yes" "Yes" ...
## $ Dependents    : int   0 1 0 0 0 2 0 3 2 1 ...
## $ Education     : chr  "Graduate" "Graduate" "Graduate" "Not
Graduate" ...
```

```

## $ Self_Employed      : chr  "No" "No" "Yes" "No" ...
## $ ApplicantIncome    : int   5849 4583 3000 2583 6000 5417 2333 3036
4006 12841 ...
## $ CoapplicantIncome: num   0 1508 0 2358 0 ...
## $ LoanAmount         : int   NA 128 66 120 141 267 95 158 168 349 ...
## $ Loan_Amount_Term   : int   360 360 360 360 360 360 360 360 360 360
...
## $ Credit_History     : int    1 1 1 1 1 1 1 0 1 1 ...
## $ Property_Area      : chr    "Urban" "Rural" "Urban" "Urban" ...
## $ Loan_Status        : chr    "Y" "N" "Y" "Y" ...

##      Loan_ID Gender Married Dependents      Education Self_Employed
ApplicantIncome
## 1 LP001002   Male      No           0      Graduate          No
5849
## 2 LP001003   Male     Yes           1      Graduate          No
4583
## 3 LP001005   Male     Yes           0      Graduate          Yes
3000
## 4 LP001006   Male     Yes           0 Not Graduate          No
2583
## 5 LP001008   Male      No           0      Graduate          No
6000
## 6 LP001011   Male     Yes           2      Graduate          Yes
5417
##      CoapplicantIncome LoanAmount Loan_Amount_Term Credit_History
Property_Area
## 1              0           NA           360           1
Urban
## 2             1508          128           360           1
Rural
## 3              0           66           360           1
Urban
## 4             2358          120           360           1
Urban
## 5              0          141           360           1
Urban
## 6             4196          267           360           1
Urban
##      Loan_Status
## 1              Y
## 2              N
## 3              Y
## 4              Y
## 5              Y
## 6              Y

```

The dataset consists of 598 cases and 13 attributes, which provide information about applicants and their loan applications. The attributes are as follows:

- **Loan_ID:** A unique application ID assigned to each applicant.
- **Gender:** The applicant's gender (Male or Female).
- **Married:** The applicant's marital status (Married or Not Married).
- **Dependents:** The number of people financially dependent on the applicant.
- **Education:** The applicant's education level (Graduate or Not Graduate).
- **Self_Employment:** Indicates whether the applicant is self-employed.
- **ApplicantIncome:** The total income of the applicant.
- **CoapplicantIncome:** The total income of the co-applicant (if any).
- **LoanAmount:** The loan amount requested by the applicant.
- **Loan_Amount_Term:** The duration of the loan in months.
- **Credit_History:** Indicates whether the applicant has a credit history.
- **Property_Area:** The type of area where the applicant resides (Urban, Semi-Urban, or Rural).
- **Loan_Status:** The outcome of the loan application (Approved or Not Approved).

In this report, we will apply machine learning techniques to predict loan approvals based on applicant information. The target variable for our analysis is "Loan_Status".

Preprocessing and Data Analysis

Missing Values and Duplicates

The dataset contains a total of 93 rows with a missing entry in at least one of the columns. Through statistics, we can intuitively get the total number of rows that have missing values and the number of missing values in each variable.

```
## [1] 93

##               total_missing  rate
## Loan_ID           0         0%
## Gender            0         0%
## Married           0         0%
## Dependents        12      2.01%
## Education          0         0%
```

```
## Self_Employed          0    0%
## ApplicantIncome        0    0%
## CoapplicantIncome       0    0%
## LoanAmount             21  3.51%
## Loan_Amount_Term        14  2.34%
## Credit_History          49  8.19%
## Property_Area           0    0%
## Loan_Status             0    0%
```

The total number of rows with missing values is 93. The columns which contain the missing values are Dependents, LoanAmount, Loan_Amount_Term, and Credit_History. To address missing values in dataset, employing mean imputation for numeric variables and mode imputation for categorical variables is a straightforward and effective approach. But before applying those process, we need to check for Missing Completely At Random (MCAR).

```
## Warning: package 'misty' was built under R version 4.4.3

## |-----|
## | misty 0.7.1 (2025-03-10) |
## | Miscellaneous Functions T. Yanagida |
## |-----|

## Warning: Cases with missing on all variables were removed from the
analysis: NA

## Little's MCAR Test
##
##      n nIncomp nPattern  chi2 df  pval
##  598      93        8 95.25 81 0.133
```

Since for p-value = 0.133, this value is greater than the conventional threshold of 0.05. We can concluded that fail to reject the null hypothesis that the data are Missing Completely At Random (MCAR). This suggests there's no strong evidence against the assumption that the missingness in data is entirely random.

Thus for each numeric column with missing values, replace the NA entries with the mean of the non-missing values in that column. And for categorical variables, replace missing values with the mode (the most frequent category).

```
## [1] 0
```

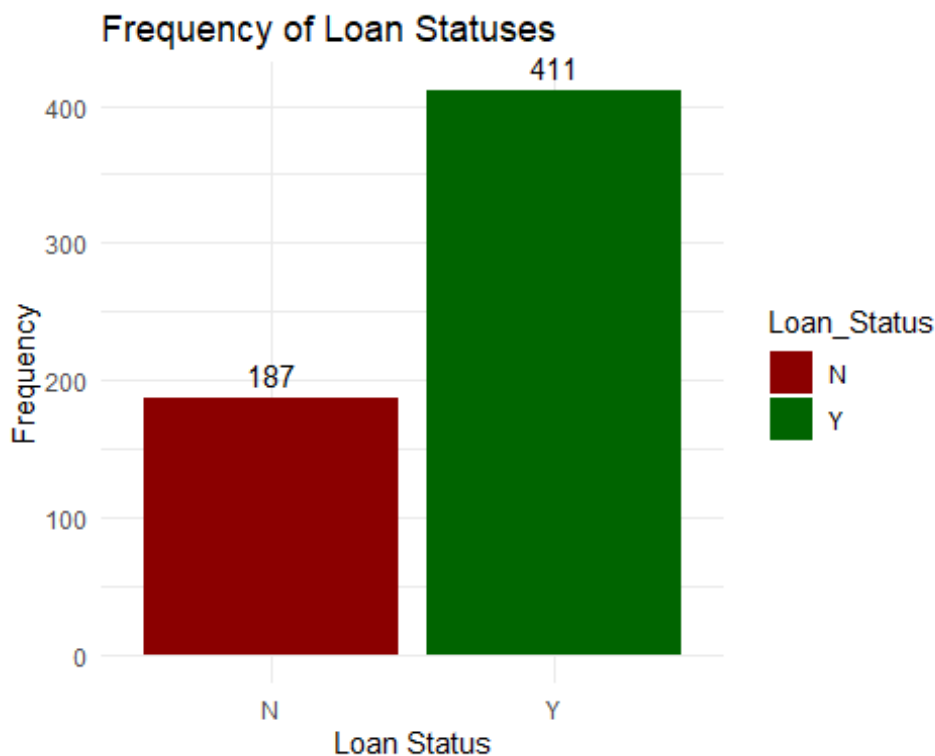
We've also assigned the proper datatype to each column and checked for any duplicates but found none. Since there are no duplicate applications, we can remove the ID variable as it will not be needed for future analysis. We now have a total of 598 rows and 12 variables in our dataset.

```
## [1] 0
```

```
## 'data.frame':    598 obs. of  12 variables:
## $ Gender          : Factor w/ 2 levels "Female","Male": 2 2 2 2 2
2 2 2 2 2 ...
## $ Married         : Factor w/ 2 levels "No","Yes": 1 2 2 2 1 2 2 2
2 2 ...
## $ Dependents      : num  0 1 0 0 0 2 0 3 2 1 ...
## $ Education       : Factor w/ 2 levels "Graduate","Not Graduate":
1 1 1 2 1 1 2 1 1 1 ...
## $ Self_Employed   : Factor w/ 2 levels "No","Yes": 1 1 2 1 1 2 1 1
1 1 ...
## $ ApplicantIncome : num  5849 4583 3000 2583 6000 ...
## $ CoapplicantIncome: num  0 1508 0 2358 0 ...
## $ LoanAmount       : num  145 128 66 120 141 267 95 158 168 349 ...
## $ Loan_Amount_Term : num  360 360 360 360 360 360 360 360 360 360
...
## $ Credit_History   : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 1 2
2 ...
## $ Property_Area    : Factor w/ 3 levels "Rural","Semiurban",...: 3 1
3 3 3 3 3 2 3 2 ...
## $ Loan_Status      : Factor w/ 2 levels "N","Y": 2 1 2 2 2 2 2 1 2
1 ...
```

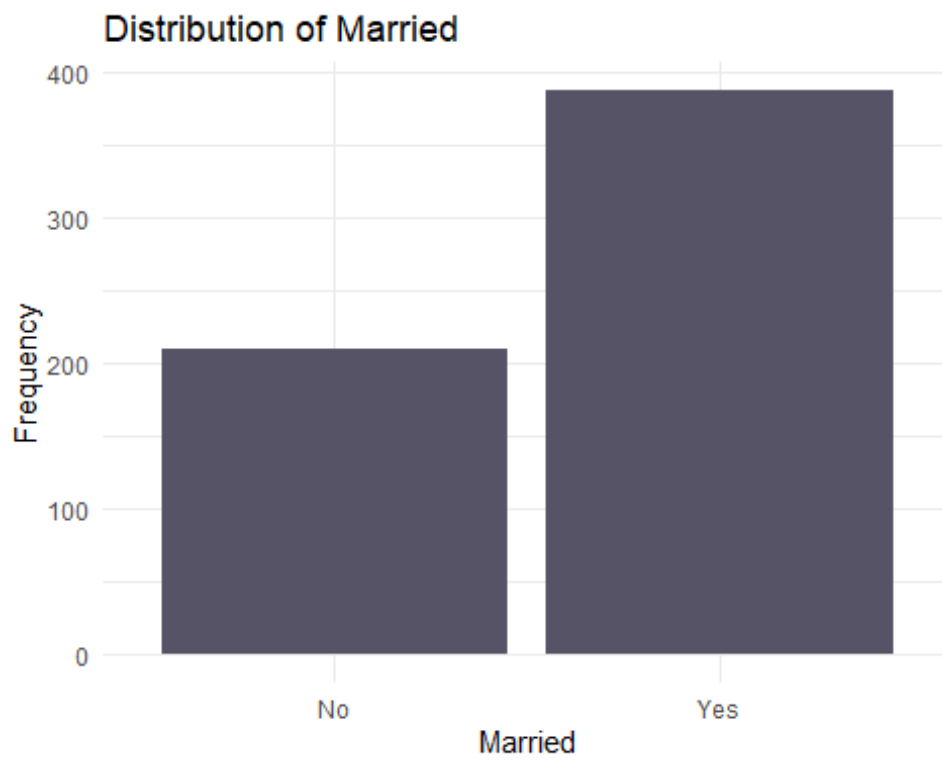
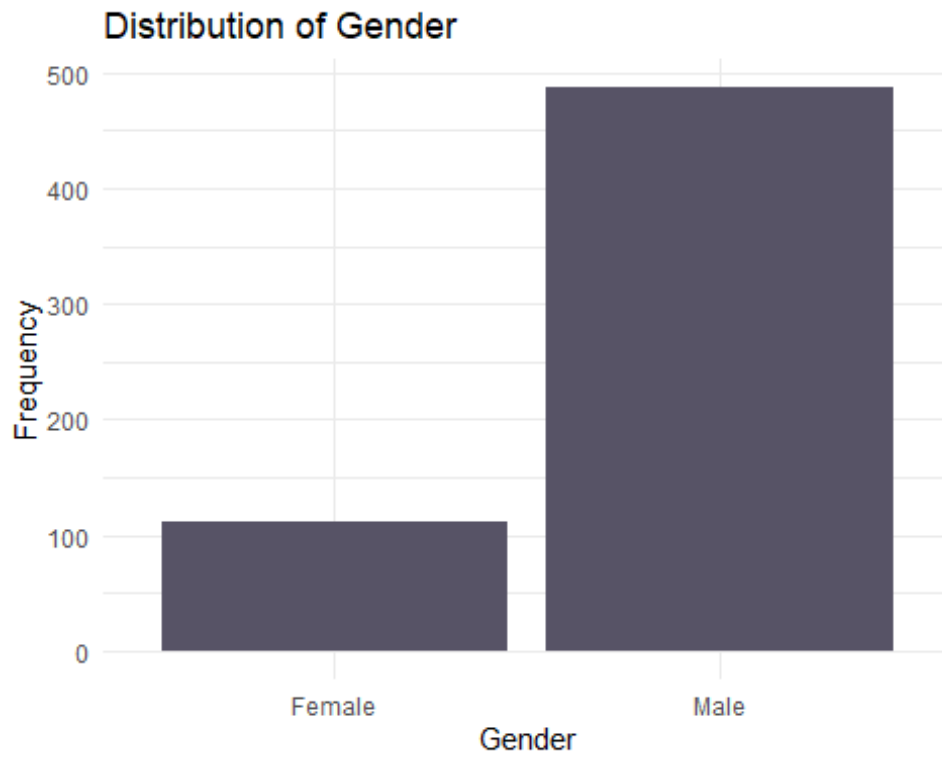
Visualization

To begin our analysis, we examine the distribution of approved and rejected loans in the dataset.

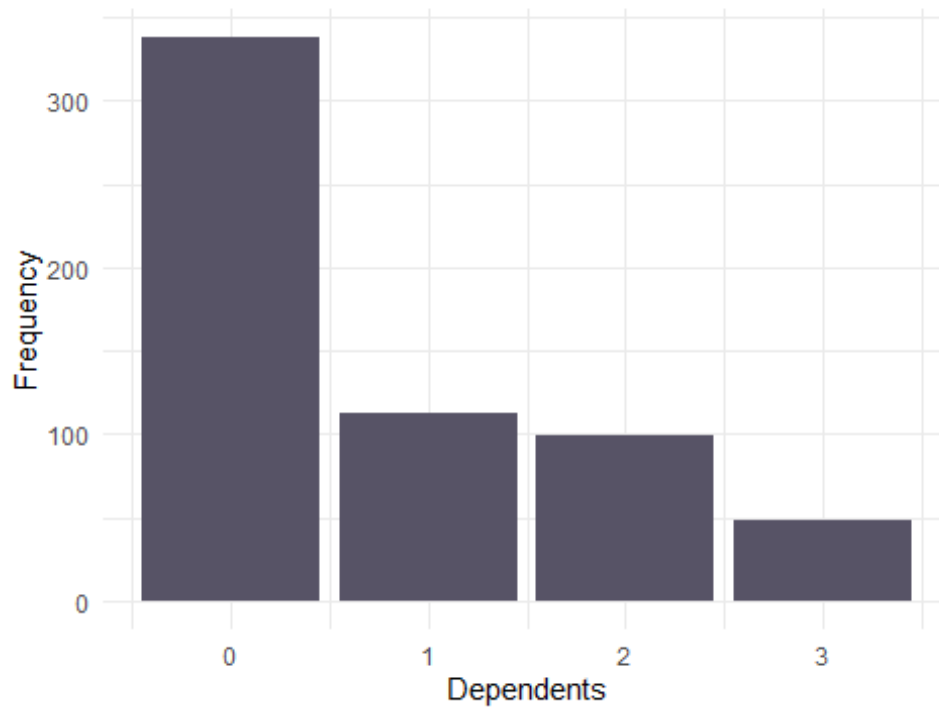


There is a clear imbalance in the number of approved versus denied loan applications. Out of the total 598 cases, 411 applicants were granted a loan, while 187 were denied.

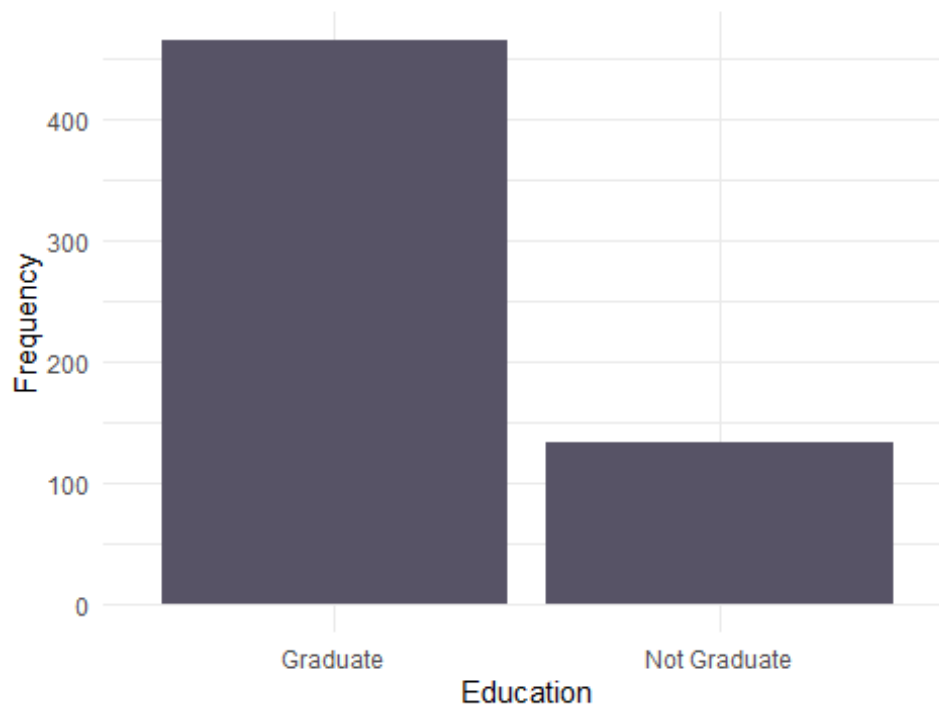
We can also examine the distribution of all other attributes in our dataset.

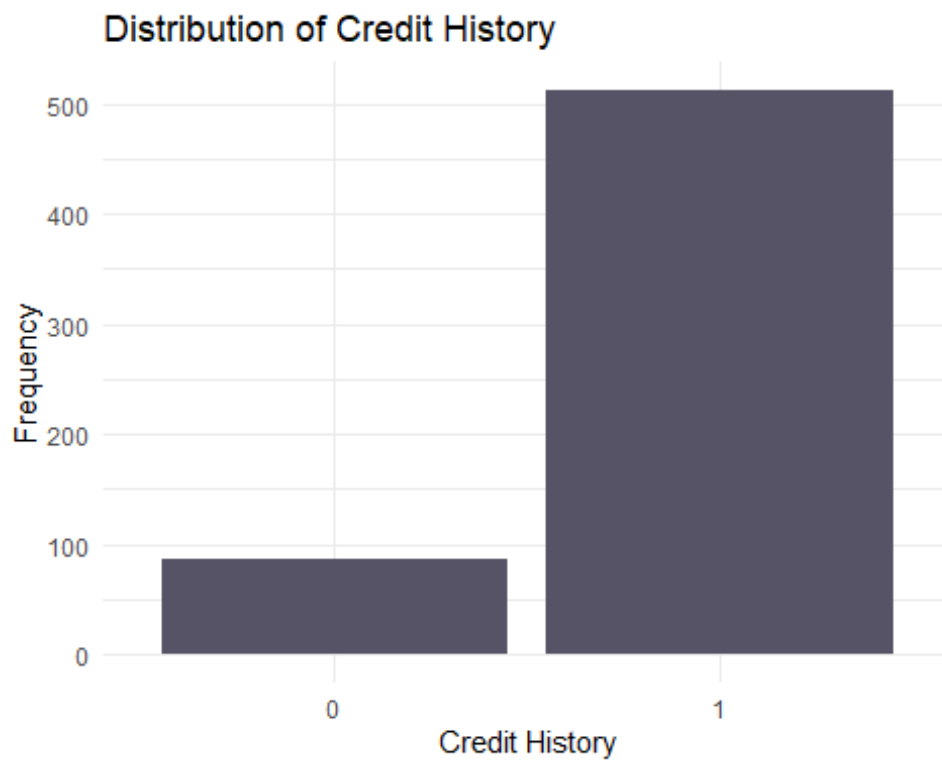
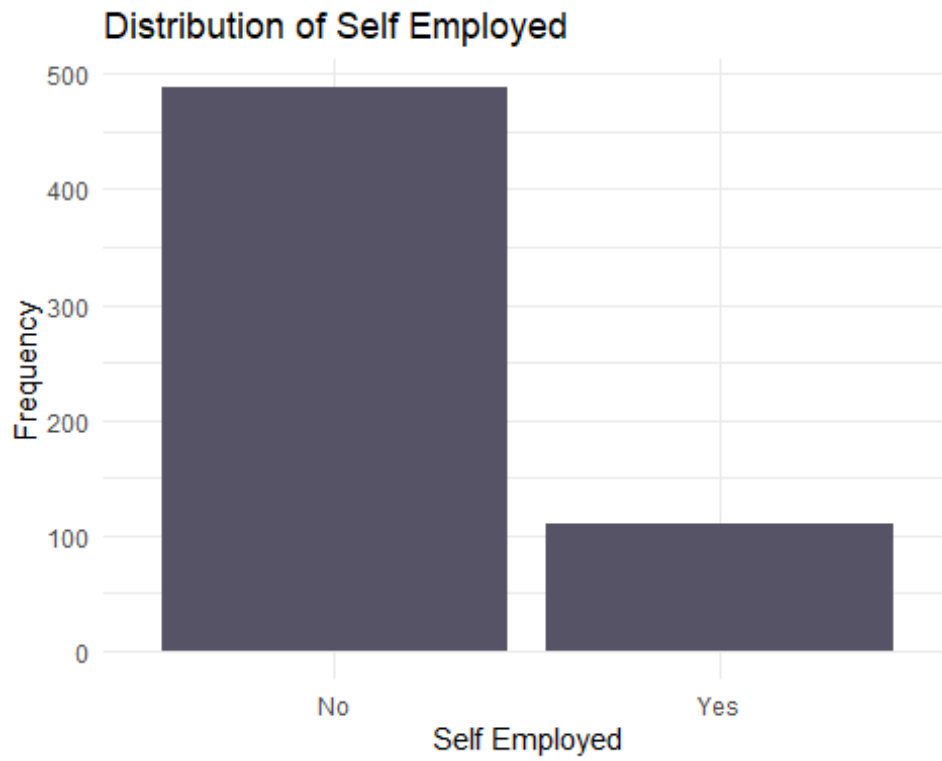


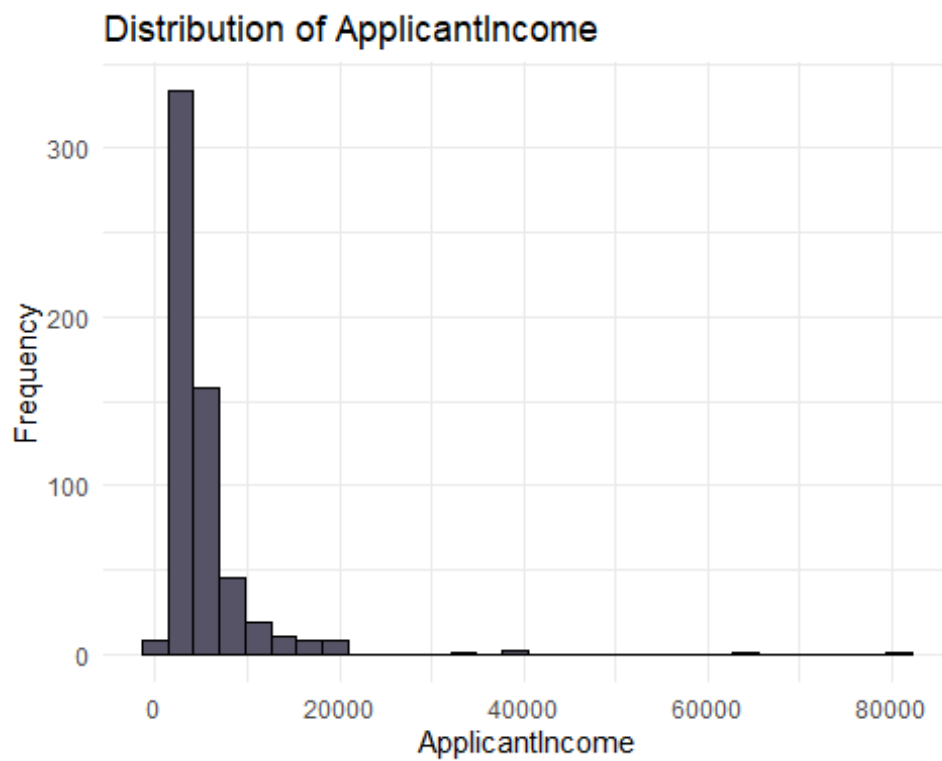
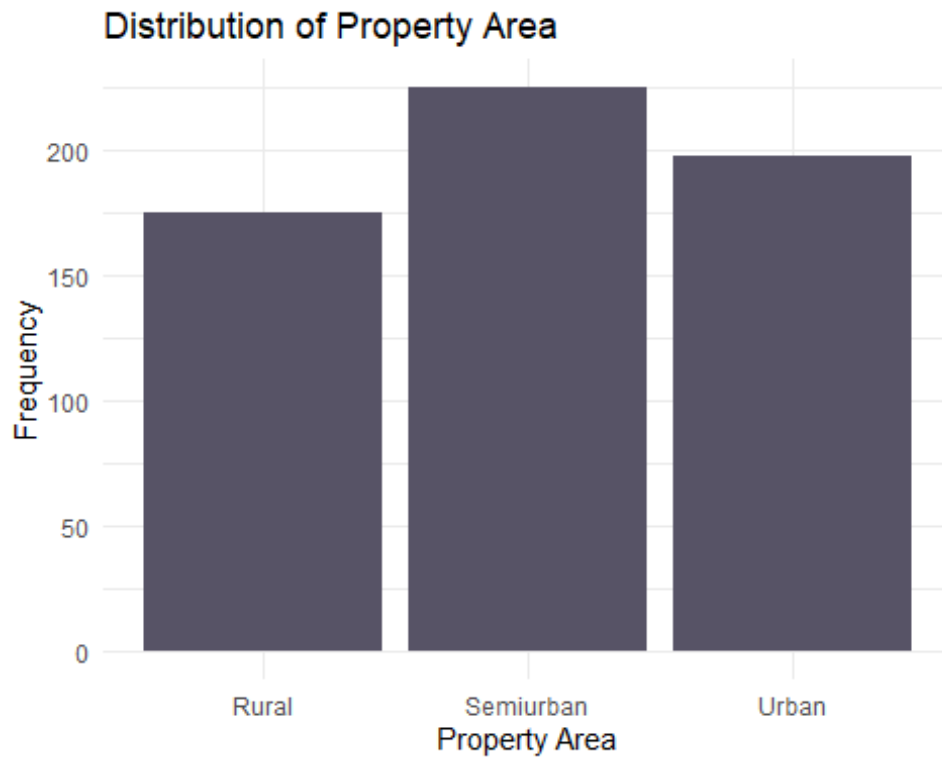
Distribution of Dependents



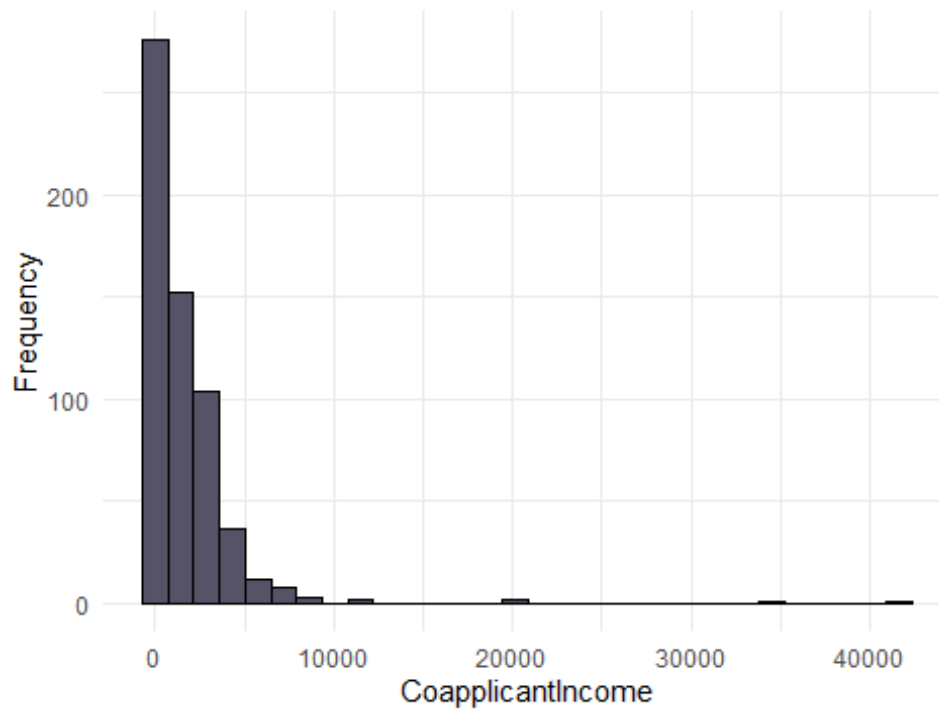
Distribution of Education



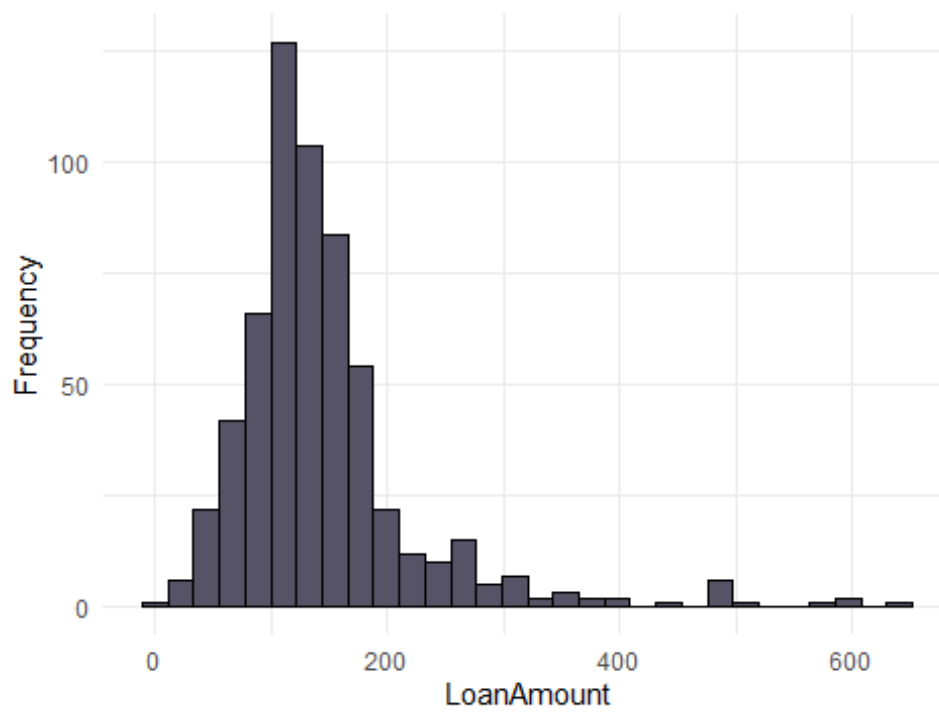


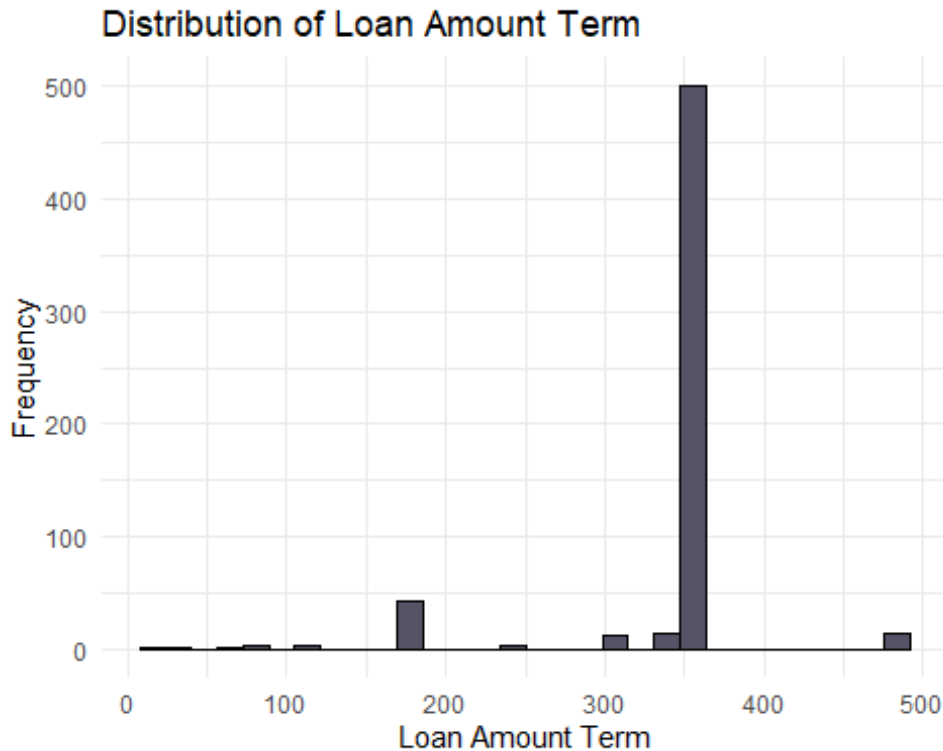


Distribution of CoapplicantIncome



Distribution of LoanAmount





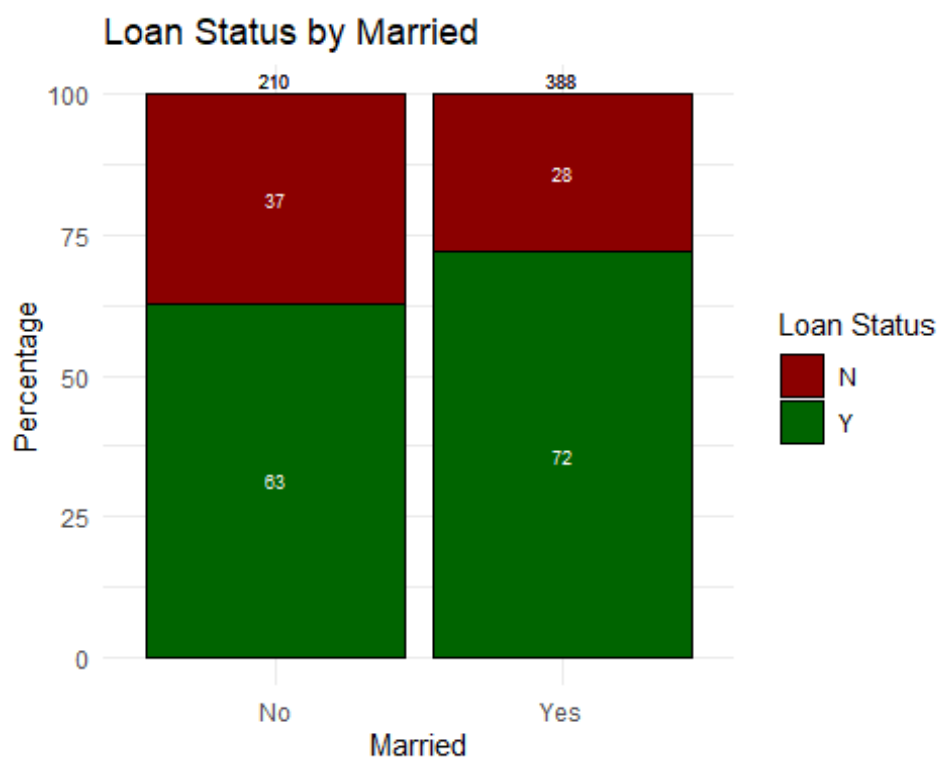
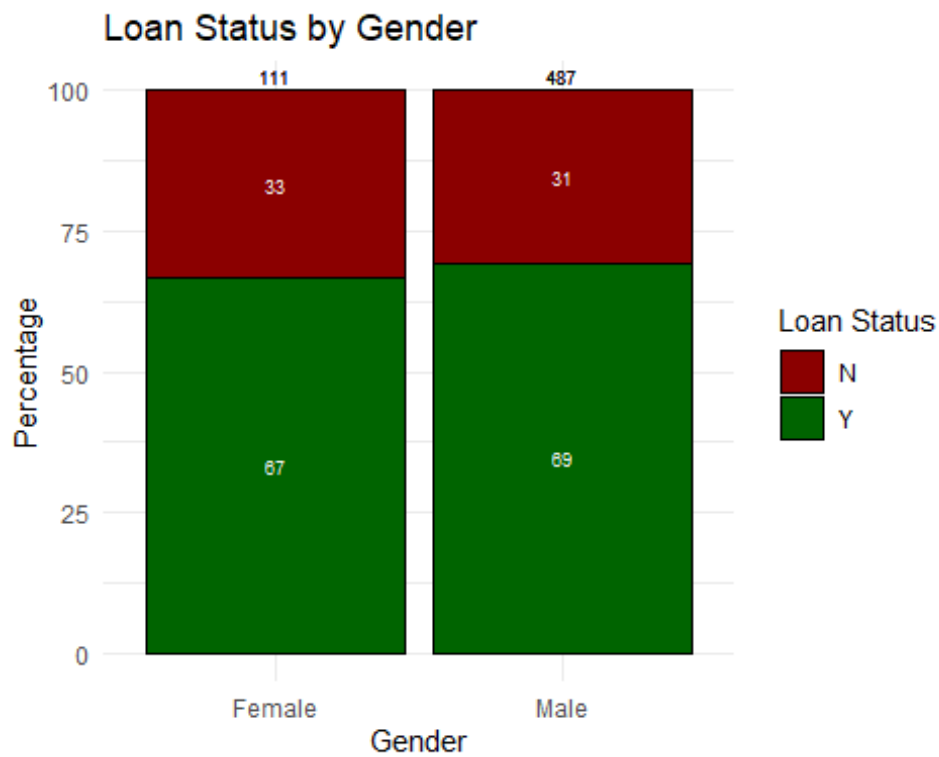
The categorical variables with two levels exhibit significant imbalances in their frequency distributions. Specifically, the dataset contains more male applicants, married individuals, graduate students, non-self-employed applicants, and individuals with a credit history. This suggests that the majority of loan applicants share certain demographic characteristics.

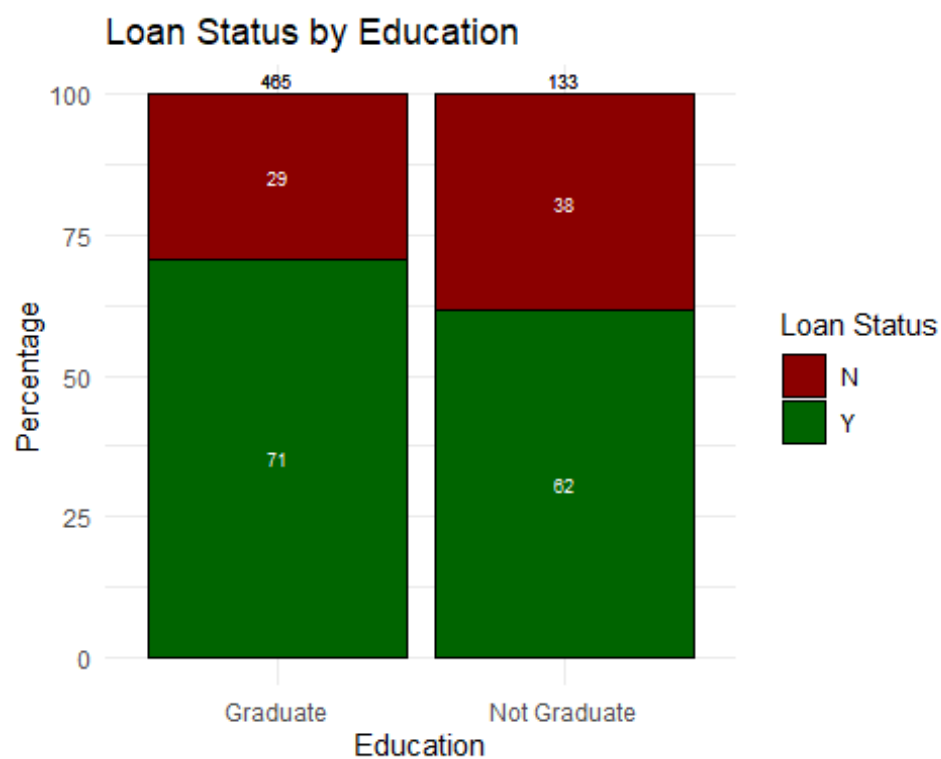
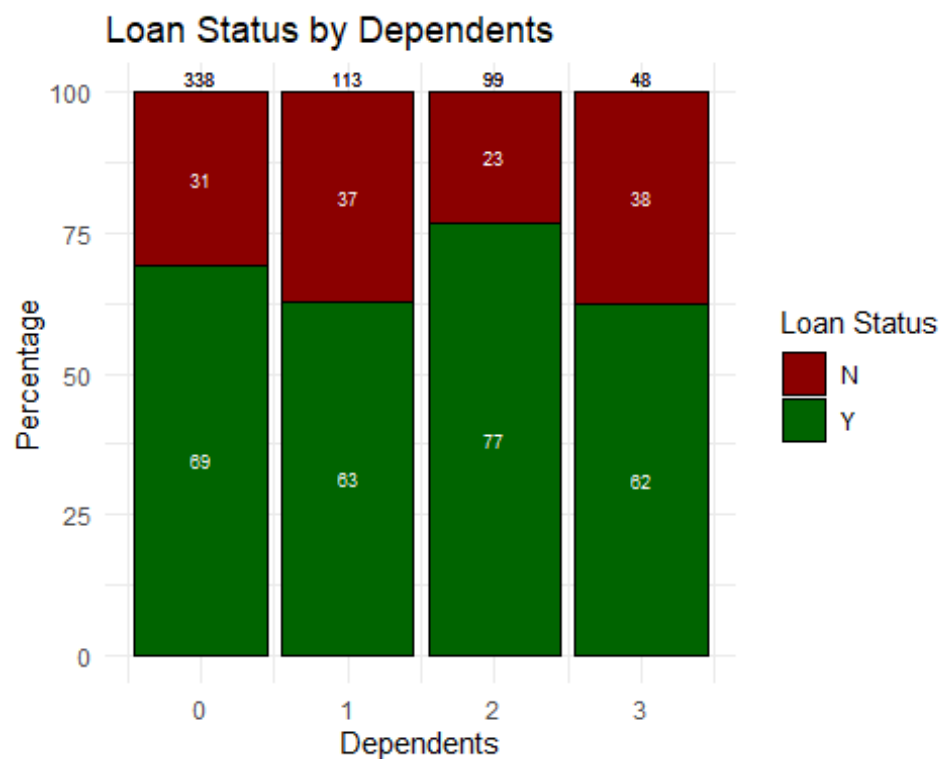
For the Property Area variable, which has three levels, the distribution shows a similar number of rural and urban residents, while a larger number of applicants live in semi-urban areas.

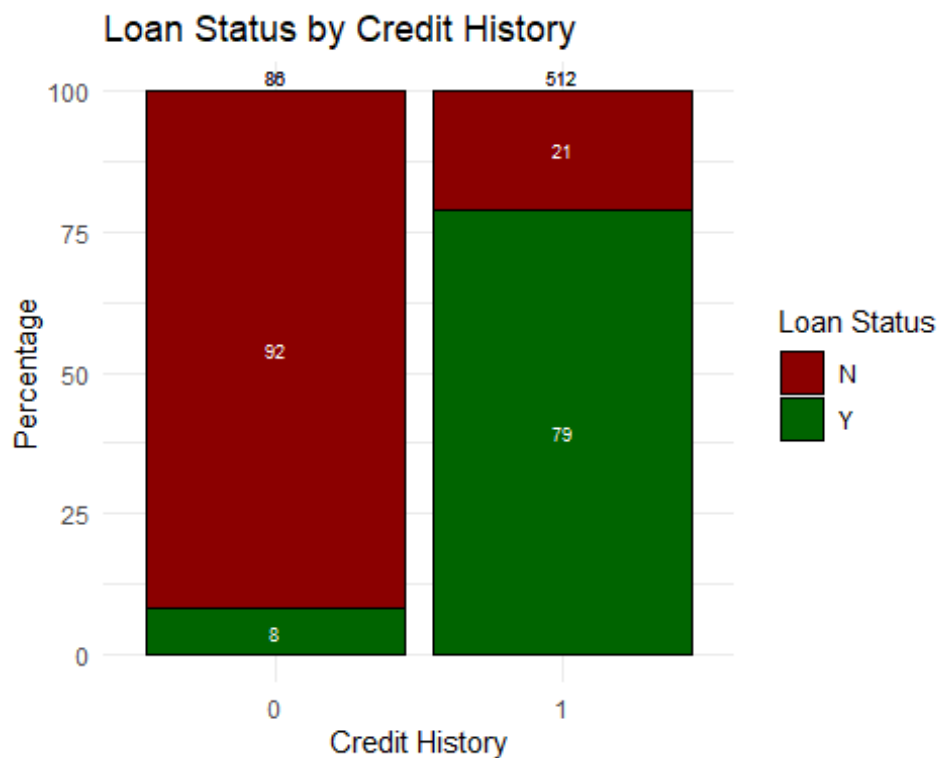
Regarding the Dependents variable, most applicants have no dependents, while the number of applicants with one to three dependents is significantly lower, indicating a right-skewed distribution.

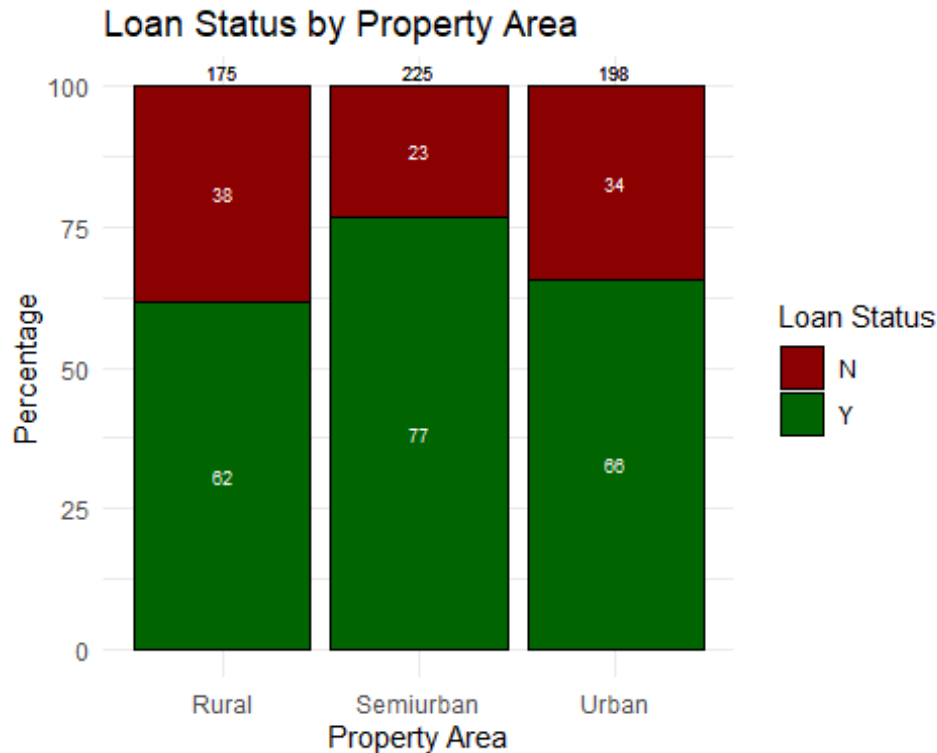
Among the continuous variables, Applicant Income, Co-applicant Income, and Loan Amount are heavily right-skewed, whereas Loan Amount Term exhibits strong left skewness.

Given the imbalance within the categorical variables and the discrete variable, rather than focusing solely on raw counts, we will analyze the proportions of loan statuses within each of those variables to better understand their relationship with loan outcomes.









Among the examined features, credit history stands out as the strongest predictor of loan approval. The “Credit History by Loan Status” plot shows that applicants with a credit history are far more likely to be approved, while those without one face a much higher likelihood of rejection. This suggests that having a credit history is a key requirement for banks to grant a loan.

Other attributes show weaker but still noticeable relationships with loan status:

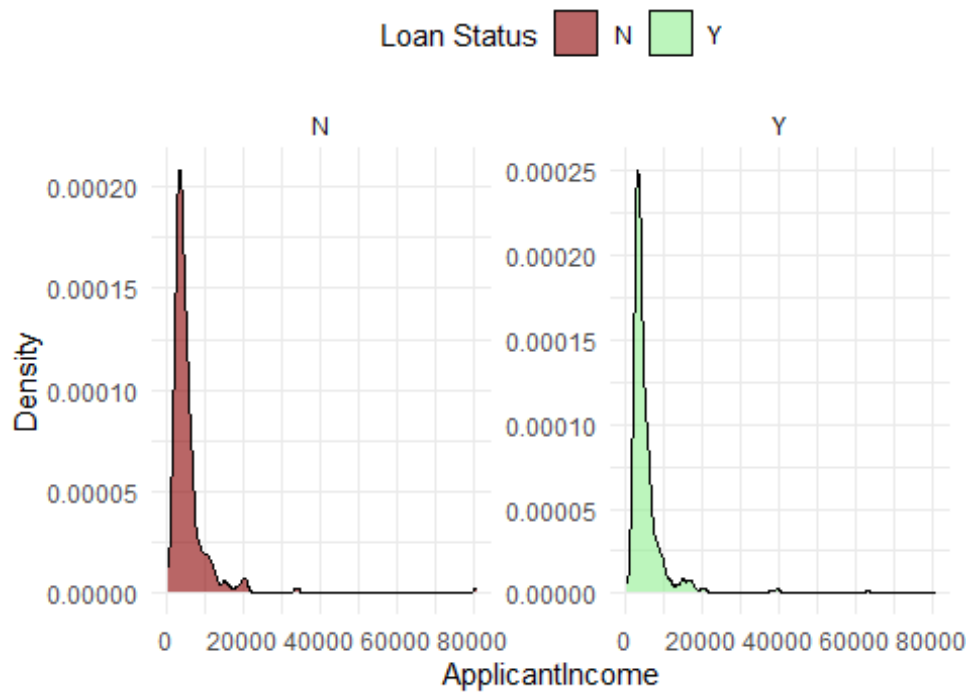
- Marital Status: Married applicants had a slightly higher approval rate than unmarried ones.
- Dependents: Applicants with two dependents saw slightly more approvals followed by those with no dependents, while those with one or three dependents had similarly lower approval rates.
- Education: Graduates were approved at a slightly higher rate than non-graduates.
- Property Area: Loan approvals were more common among applicants from semi-urban areas, while those from urban and rural areas experienced higher rejection rates in comparison.

Finally, self-employment status and gender have no apparent impact on loan approval, suggesting that banks do not weigh these factors significantly when making lending decisions.

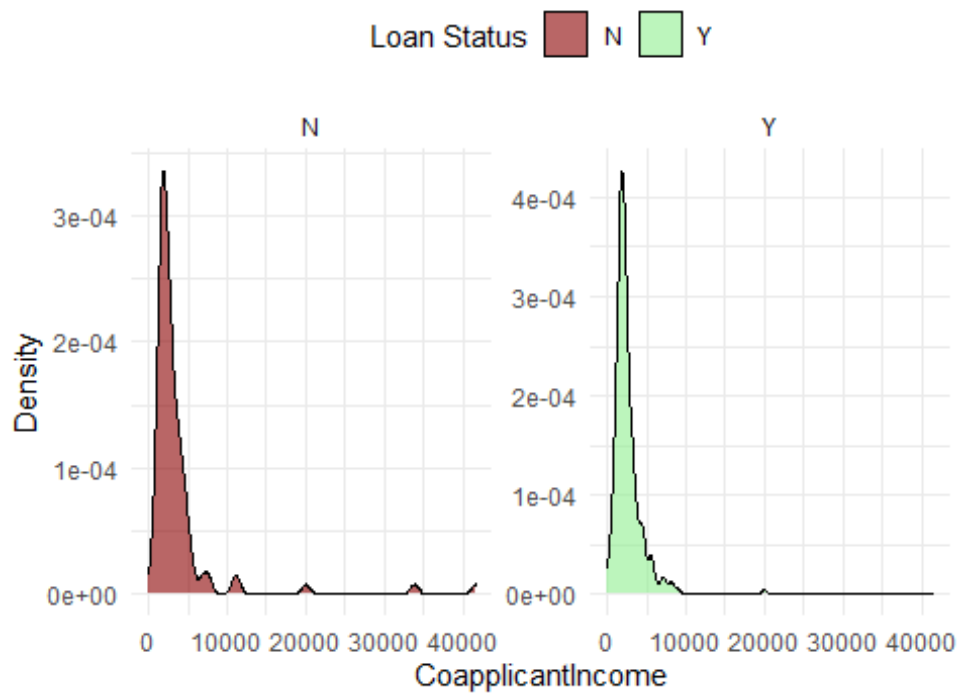
For continuous variables, density plots were generated to compare the distributions of approved and rejected applicants.

Before visualizing, zero values were excluded from co-applicant income, as co-applicants are typically intended to provide financial support when the primary applicant alone may not qualify for the loan. A value of zero likely indicates that no co-applicant was involved, rather than an actual absence of income. Since co-applicants are generally required for low-income applicants or those with insufficient credit history, removing these zeros ensures a more accurate representation in the density plots.

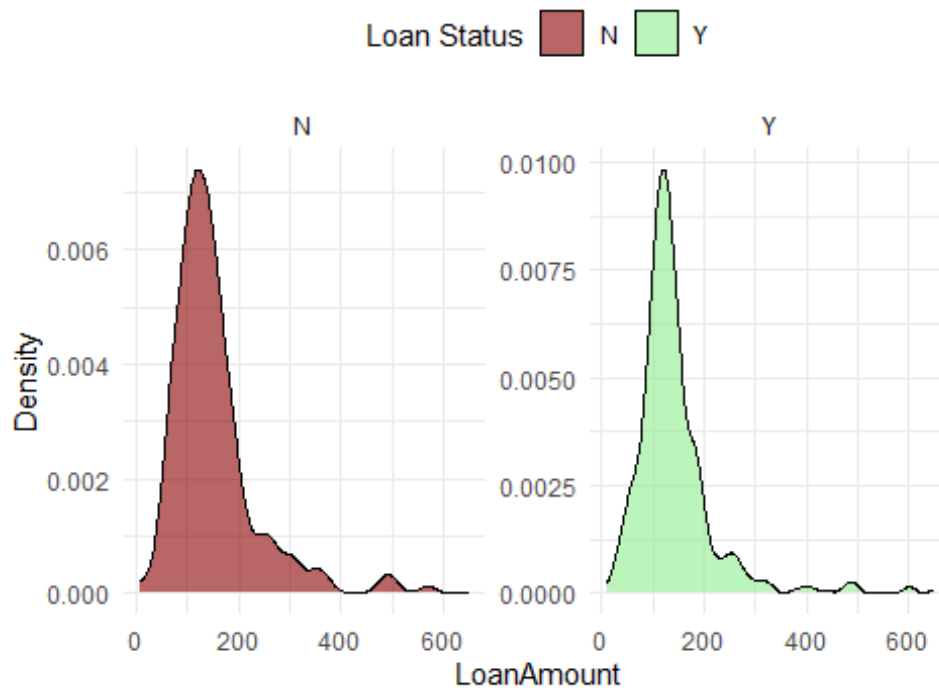
Density Plot of ApplicantIncome by Loan Status



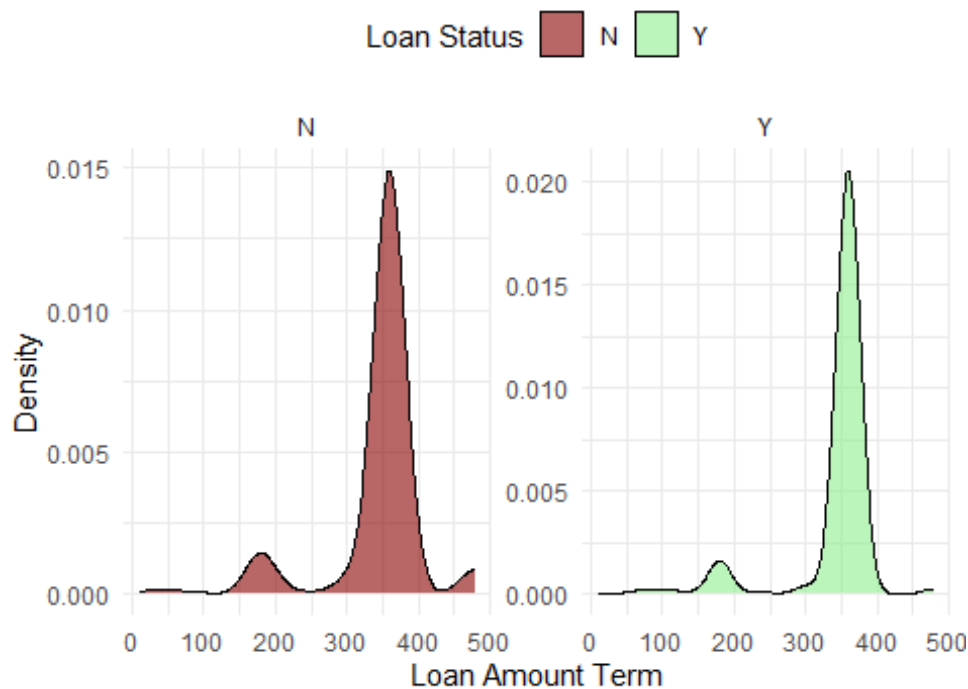
Density Plot of CoapplicantIncome by Loan Status



Density Plot of LoanAmount by Loan Status



Density Plot of Loan Amount Term by Loan Status



The density plots show that the distributions for approved and rejected applicants are largely similar, with both peaking around the same values. Although the curves are similar for both loan statuses, there is a subtle but meaningful distinction. The

density curve for approved applicants appears slimmer, indicating a higher concentration around the mode, whereas the rejected applicants exhibit greater variability. This may suggest that banks approve loan applications when these continuous variables fall within a preferred range.

Encoding Categorical Variables

The dataset contains several binary variables, each with two categories. These were encoded using binary representation, where values were transformed into 0s and 1s. For the variable Property Area, which includes three categories, ordinal encoding was applied by assigning numeric values from 0 to 2.

Encoding the categorical variables with two levels into binary form allows for the application of logistic Principal Component Analysis (logistic PCA) later in the analysis. Logistic PCA is a dimensionality reduction technique similar to standard PCA but specifically designed for binary data.

The variable encodings are as follows:

- Gender: 1 = Male, 0 = Female
- Married: 1 = Married, 0 = Not Married
- Education: 1 = Graduate, 0 = Not Graduate
- Self_Employed: 1 = Self-employed, 0 = Not Self-employed
- Property_Area: 0 = Urban, 1 = Semi-urban, 2 = Rural
- Credit_History: 1 = Has a Credit History, 0 = No Credit History
- Loan_Status: 1 = Approved, 0 = Denied

```
## 'data.frame':    598 obs. of  12 variables:
## $ Gender          : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2
## 2 ...
## $ Married         : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 2 2
## 2 ...
## $ Dependents      : num  0 1 0 0 0 2 0 3 2 1 ...
## $ Education       : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 2
## 2 ...
## $ Self_Employed   : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 1 1
## 1 ...
## $ ApplicantIncome : num  5849 4583 3000 2583 6000 ...
## $ CoapplicantIncome: num  0 1508 0 2358 0 ...
## $ LoanAmount      : num  145 128 66 120 141 267 95 158 168 349 ...
## $ Loan_Amount_Term : num  360 360 360 360 360 360 360 360 360 360
## ...
## $ Credit_History  : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 1 2
```

```

2 ...
## $ Property_Area      : Factor w/ 3 levels "0","1","2": 1 3 1 1 1 1 1
2 1 2 ...
## $ Loan_Status        : Factor w/ 2 levels "0","1": 2 1 2 2 2 2 2 1 2
1 ...

```

Correlation

To assess multicollinearity and redundancy among numeric variables, we computed pairwise Pearson correlations between numerical variables (Dependents, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term).

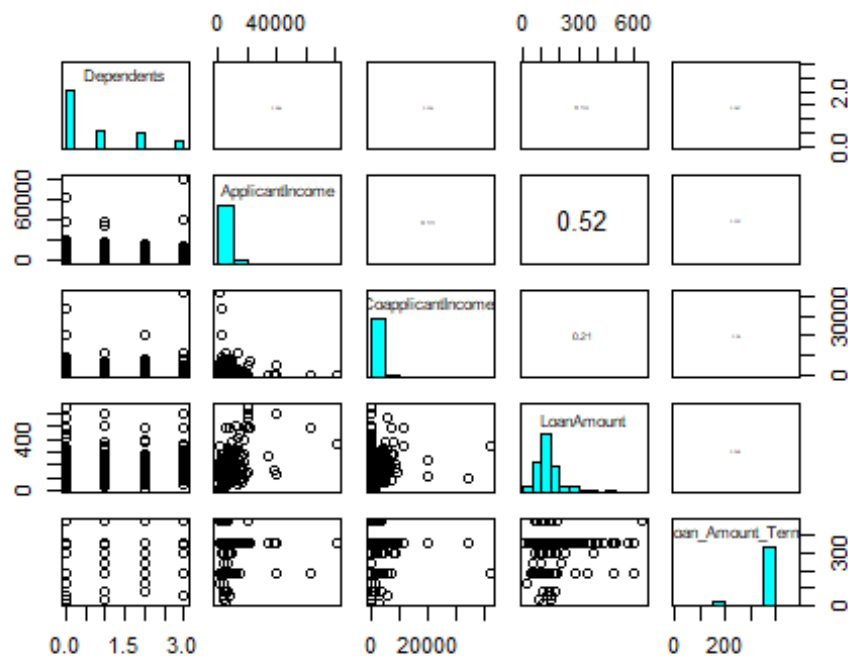
```

##               Dependents ApplicantIncome CoapplicantIncome
LoanAmount
## Dependents      1.00000000      0.08392527      0.03390232
0.12703630
## ApplicantIncome  0.08392527      1.00000000     -0.10923451
0.52468130
## CoapplicantIncome 0.03390232     -0.10923451      1.00000000
0.20621581
## LoanAmount      0.12703630      0.52468130      0.20621581
1.00000000
## Loan_Amount_Term -0.09749143     -0.03534045     -0.06019726
0.04858012
##               Loan_Amount_Term
## Dependents      -0.09749143
## ApplicantIncome  -0.03534045
## CoapplicantIncome -0.06019726
## LoanAmount       0.04858012
## Loan_Amount_Term 1.00000000

```

The strongest correlation is observed between ApplicantIncome and LoanAmount ($r \approx 0.52$), which is moderate and expected, as higher-income applicants tend to request larger loans.

All other variable pairs exhibit weak correlations ($|r| < 0.2$), suggesting that they provide largely independent information.



Notably, CoapplicantIncome shows little to no correlation with the other variables, including LoanAmount, indicating it may carry distinct predictive value.

Overall, the low correlation among most numeric variables supports their inclusion in modeling and dimensionality reduction without significant concern for multicollinearity.

Split data

When preparing data for building a classification, it's essential to split the dataset into training and testing subsets to evaluate the model's performance accurately.

The total amount of processed data is 598 obs., we divide it into training data and test data in a ratio of 80% to 20% (commonly used).

```
## Warning: package 'caTools' was built under R version 4.4.3

##
##      0      1
## 150 329

##
##      0      1
## 37  82
```

The training set contains 479 observation in total with 329 approved cases and 150 denied cases. The test set contains 119 total observations in total with 82 approved cases and 37 denied cases.

Undersampling

The original dataset exhibited a noticeable imbalance between approved (Loan_Status = 1) and denied (Loan_Status = 0) loan applications, with approvals comprising a significantly larger proportion. This imbalance can bias supervised learning models toward the majority class, reducing their ability to correctly identify minority-class instances.

To address this, we applied random undersampling to the majority class in the training set, resulting in a balanced dataset with an equal number of approved and denied cases.

Also, undersampling was not applied to the test data in order to preserve its original class distribution, ensuring that model evaluation reflects real-world performance on naturally imbalanced data.

```
##
##    0    1
## 150 150
```

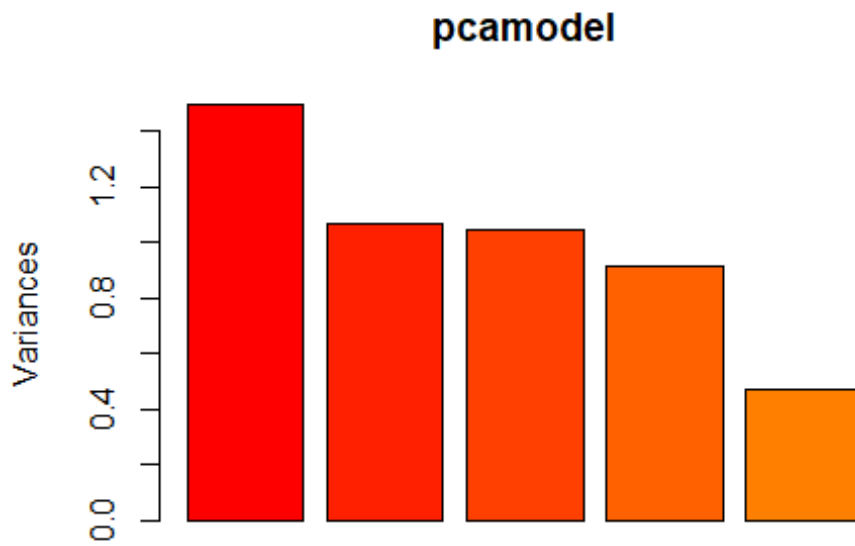
Dimension Reduction

To reduce the dimensionality of the dataset while preserving most of the variance in the numeric features, we applied Principal Component Analysis (PCA) exclusively to the numeric variables (Dependents, ApplicantIncome, CoapplicantIncome, LoanAmount, and Loan_Amount_Term).

Standard PCA requires continuous, numerical input and does not offer meaningful interpretations for categorical values, even if encoded as numbers, since the ordinal structure of such data may be arbitrary or misleading.

```
## Importance of components:
##
##          PC1      PC2      PC3      PC4      PC5
## Standard deviation    1.2232  1.0350  1.0228  0.9565  0.68670
## Proportion of Variance 0.2992  0.2142  0.2092  0.1830  0.09431
## Cumulative Proportion 0.2992  0.5135  0.7227  0.9057  1.00000
```

The resulting components show that the first principal component (PC1) accounts for 29.9% of the total variance, and the first four components together capture over 90.6% of the total variance. This justifies the use of a reduced set of principal components in downstream analysis, as they retain most of the informative variability of the numeric data while minimizing redundancy.



We retained the first four principal components, which together capture approximately 90.6% of the total variance in the numeric variables. This strikes a balance between dimensionality reduction and information retention, while excluding the fifth component, which contributes minimal additional variance.

##	PC1	PC2	PC3	PC4
## Dependents	0.24355101	0.004539983	0.62672877	0.73872644
## ApplicantIncome	0.61695518	0.406443463	0.05574946	-0.29124317
## CoapplicantIncome	0.23073650	-0.870254279	-0.17878856	0.05626677
## LoanAmount	0.70603822	-0.105587270	-0.10979472	-0.09592001
## Loan_Amount_Term	0.09121356	0.257472170	-0.74838681	0.59757349

PC1 = 0.71 x LoanAmount + 0.62 x ApplicantIncome + 0.24 x (Dependents + CoapplicantIncome) + 0.09 x Loan_Amount_Term

- PC1 captures the combined effect of the loan amount and the applicant's income.

PC2 = - 0.87 x CoapplicantIncome + 0.41 x ApplicantIncome + 0.26 x Loan_Amount_Term - 0.11 x LoanAmount

- PC2 captures the effect of the co-applicant income with some contrast from the applicant income.
- Hence, it is comparing how much the coapplicant earns against the salary of the applicant.

$PC3 = -0.75 \times \text{Loan_Amount_Term} + 0.63 \times \text{Dependents} - 0.18 \times \text{CoapplicantIncome} - 0.11 \times \text{LoanAmount} + 0.06 \times \text{ApplicantIncome}$

- PC3 captures the effect of the loan amount term versus the number of dependents.
- Hence, it is comparing how long the applicant will be granted the loan against how many people depend on them financially

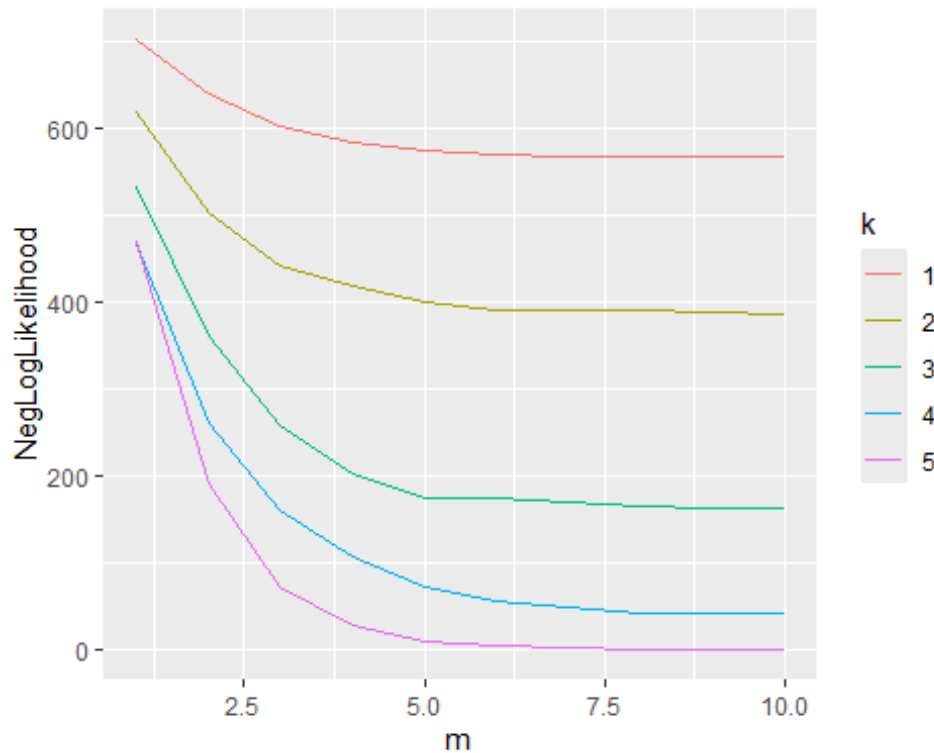
$PC4 = 0.74 \times \text{Dependents} + 0.60 \times \text{Loan_Amount_Term} - 0.29 \times \text{ApplicantIncome} - 0.10 \times \text{LoanAmount} + 0.06 \times \text{CoapplicantIncome}$

- PC4 captures the combined effect of the number of dependents and the length of the loan.
- PC3 is the contrast between the length of the loan and the number of dependents meanwhile PC4 is their merged effect.

For the binary variables, we applied Logistic Principal Component Analysis (Logistic PCA) to perform dimensionality reduction on it. Standard PCA is not suitable for binary data due to its assumption of continuous variables. The principal components for the binary variables will be referred to as PCB and the principal components for the numerical variables will be referred to as PC.

Since Logistic PCA is only applicable to binary variables, we will remove Property_Area because it is ternary. The target variable is also not included.

```
## Warning: package 'logisticPCA' was built under R version 4.4.3
## Warning in type.convert.default(colnames(x)): 'as.is' should be
specified by
## the caller; using TRUE
## Warning in type.convert.default(rownames(x)): 'as.is' should be
specified by
## the caller; using TRUE
```



We determined the optimal number of principal components is $k=4$ and the saturation parameter is $m=5$ through cross-validation, selecting the combination that minimized the negative log-likelihood.

```
## 300 rows and 5 columns
## Rank 4 solution with m = 5
##
## 93.6% of deviance explained
## 71 iterations to converge
```

A logistic PCA was performed on the binary categorical variables, selecting $k = 4$ components. This configuration explained 94% of the deviance, and shows a large amount of data variability is captured while keeping the model simple. The choice balances the trade-off between model complexity and explanation.

PCB1 = - 0.54 x Gender + 0.45 x (Self_Employed - [Education + Credit_History]) - 0.32 x Married

- All the variables have a high contribution
- The highest contributor is the gender. Hence, PCB1 can be consider to mainly represent gender.
- There is also significant contribution from self-employment status, educational status and presence of credit history, where self-employment status is contrasted with the other two.

$PCB2 = -0.73 \times \text{Married} + 0.41 \times \text{Credit_History} - 0.38 \times \text{Gender} + 0.33 \times \text{Education} - 0.23 \times \text{Self_Employed}$

- The main contributor is marital status with some comparison to the presence of a credit history.
- Hence, PCB2 mainly represents the difference between the applicant's gender and their credit history.

$PCB3 = -0.67 \times \text{Education} + 0.53 \times \text{Gender} - 0.44 \times \text{Married} - 0.23 \times \text{Self_Employed} + 0.13 \times \text{Credit_History}$

- The two largest negative contributors are the education status and marital status.
- The largest positive contributor is gender.
- Thus, PCB3 is represented by the compared effect of educational and marital status against gender.

$PCB4 = 0.71 \times \text{Self_Employed} + 0.69 \times \text{Credit_History} + 0.09 \times (\text{Gender} - \text{Education}) + 0.07 \times \text{Married}$

- The dominating variables are the self-employment status and the presence of a credit history.
- Therefore, PCB4 represents largely the combined effect of an applicant's self-employment status and their credit history.

After performing PCA, the numerical and categorical results were combined into a single dataset along with the target variable and the variable Property Area. This new dataset is the training set that will be used for later analysis and modeling.

The same measures are applied to the test data sets to ensure consistency. That is, the logistic and standard PCAs obtained from the training set were applied to the test set.

Machine Learning

Unsupervised Learning

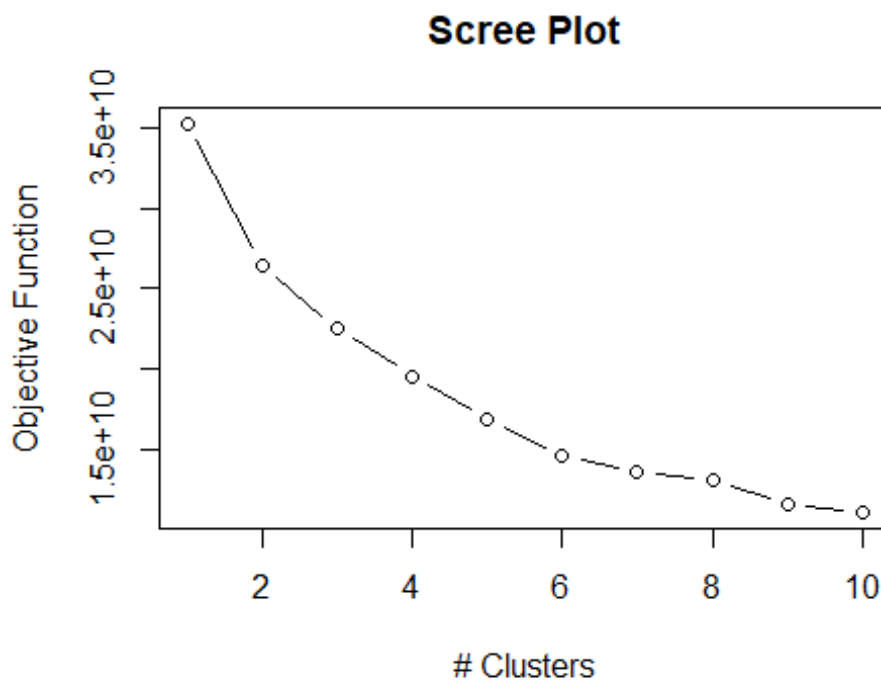
For this section, we will perform clustering analysis which is an unsupervised learning method in order to group individuals with similar loan applications into clusters. Two clustering methods will be used: k-Prototypes and Gower Distance. Unsupervised learning differs from supervised learning since it attempts to discover patterns in the data without knowing the outcome. For this reason we will work with the raw data as it is not required to use a training and tests set.

k-Prototypes

The first clustering method we will use is k-Prototypes clustering which can be applied to mixed data. K-prototypes is heavily influenced by the majority classes since centroids are biased towards majority values. Consequently, we've undersampled the loan dataset to balance the number of approved and not approved loan applications in order to use the k-prototype clustering method.

```
##  
##  0  1  
## 187 187
```

Once balanced, the data contained 187 observations for each the approved loans and the denied loans. The target variable was then removed from the dataset. To start, we first we need to determine the optimal number of clusters. We can do so by using the standard scree test.



The scree plot of the objective function E reveals a substantial drop from $k=1$ to $k=2$ clusters. There is a slight decrease in the objective function E from $k=2$ to $k=5$. Beyond $k=5$, the rate of decline becomes less impactful, indicating minimized returns in model improvement. Using the elbow method, the optimal number of clusters, where no stronger improvements can be seen is $k = 3$.

Therefore, we select $k=3$ clusters for our analysis.

In applying the kproto function to our balanced data, we want to choose nstart to be large enough to give us a better chance of finding global optimum.

```
## Distance type: huang
##
## Numeric predictors: 5
## Categorical predictors: 6
## Lambda: 24708827
##
## Number of Clusters: 3
## Cluster sizes: 3 135 236
## Within cluster error: 1233950303 8573050695 12788523487
##
## Cluster prototypes:
##   Gender Married Dependents Education Self_Employed ApplicantIncome
## 1      1      1      1.3333333      1      0      59394.333
## 2      1      0      0.4888889      1      0      6734.993
## 3      1      1      0.9279661      1      0      3872.114
##   CoapplicantIncome LoanAmount Loan_Amount_Term Credit_History
Property_Area
## 1      0.0000      370.000      300.0000      1
0
## 2      687.5333      152.200      347.7778      1
0
## 3      2298.3856      141.089      342.6864      1
1
```

The results correctly indicate there are 5 numeric predictors and 6 categorical predictors. It also lists the sizes of each cluster which is the number of points in a cluster and the within cluster error which is the sum of squared distances between each point in a cluster and the cluster's centroid.

Cluster 1 has very few points and the smallest within cluster error. Cluster 3 has a very large number of points and the largest within cluster error.

```
##
##      0      1
## 1      2      1
## 2      75     60
## 3     110    126

##
##      0      1
## 1 0.6666667 0.3333333
## 2 0.5555556 0.4444444
## 3 0.4661017 0.5338983
```

The distribution of loan status within each cluster is alarming. Although there is a dominating loan status within each cluster, the split is almost 50-50.

If we consider the majority class within each cluster, we can assign cluster 1 and 2 to denied loan applications and cluster 3 to approved loan applications.

Let's view the summary and the visualization for more details on how much proportion of each feature the clusters represents:

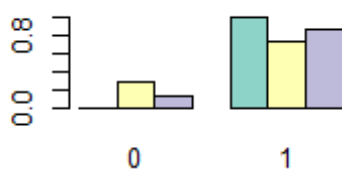
```
## Gender
##
## cluster      0      1
##      1 0.000 1.000
##      2 0.281 0.719
##      3 0.131 0.869
##
## -----
## Married
##
## cluster      0      1
##      1 0.000 1.000
##      2 0.733 0.267
##      3 0.165 0.835
##
## -----
## Dependents
##   Min. 1st Qu. Median      Mean 3rd Qu.  Max.
## 1     0     0.5      1 1.3333333      2     3
## 2     0     0.0      0 0.4888889      1     3
## 3     0     0.0      1 0.9279661      2     3
##
## -----
## Education
##
## cluster      0      1
##      1 0.000 1.000
##      2 0.200 0.800
##      3 0.254 0.746
##
## -----
## Self_Employed
##
## cluster      0      1
##      1 0.667 0.333
##      2 0.778 0.222
##      3 0.860 0.140
##
## -----
## ApplicantIncome
##   Min. 1st Qu. Median      Mean 3rd Qu.  Max.
## 1 33846 48591.50 63337 59394.333 72168.5 81000
## 2  1442  3366.50  5250  6734.993  8774.5 20833
## 3   150  2606.75  3511  3872.114  4689.0 12841
```

```

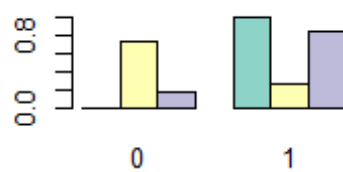
##
## -----
## CoapplicantIncome
##   Min. 1st Qu. Median      Mean 3rd Qu.  Max.
## 1    0      0    0.0    0.0000    0.00    0
## 2    0      0    0.0  687.5333  507.00  6667
## 3    0      0 1737.5 2298.3856 2791.25 41667
##
## -----
## LoanAmount
##   Min. 1st Qu. Median      Mean 3rd Qu.  Max.
## 1  260  310.00    360 370.0000  425.00  490
## 2    9   95.50    132 152.200   175.50  570
## 3   17  104.75    131 141.089   167.25  495
##
## -----
## Loan_Amount_Term
##   Min. 1st Qu. Median      Mean 3rd Qu.  Max.
## 1  180    270    360 300.0000    360   360
## 2   60    360    360 347.7778    360  480
## 3   36    360    360 342.6864    360  480
##
## -----
## Credit_History
##
## cluster    0    1
##      1 0.333 0.667
##      2 0.200 0.800
##      3 0.220 0.780
##
## -----
## Property_Area
##
## cluster    0    1    2
##      1 0.333 0.333 0.333
##      2 0.526 0.119 0.356
##      3 0.229 0.492 0.280
##
## -----

```

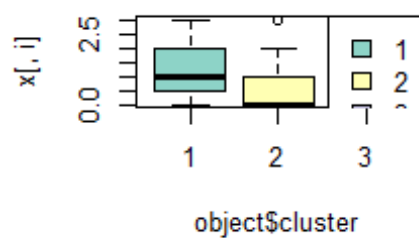
Gender



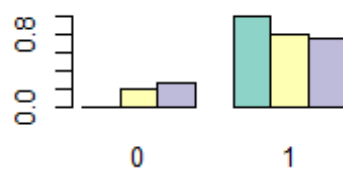
Married



Dependents



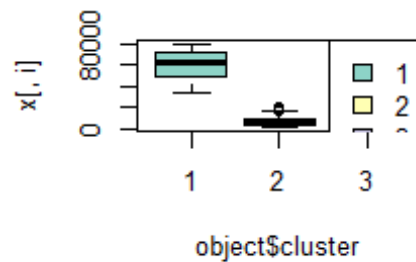
Education



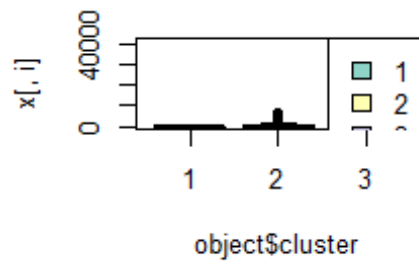
Self_Employed



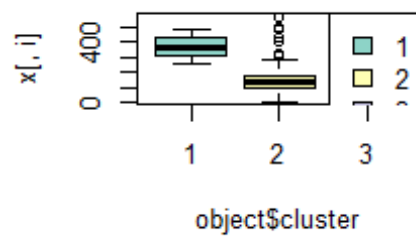
ApplicantIncome

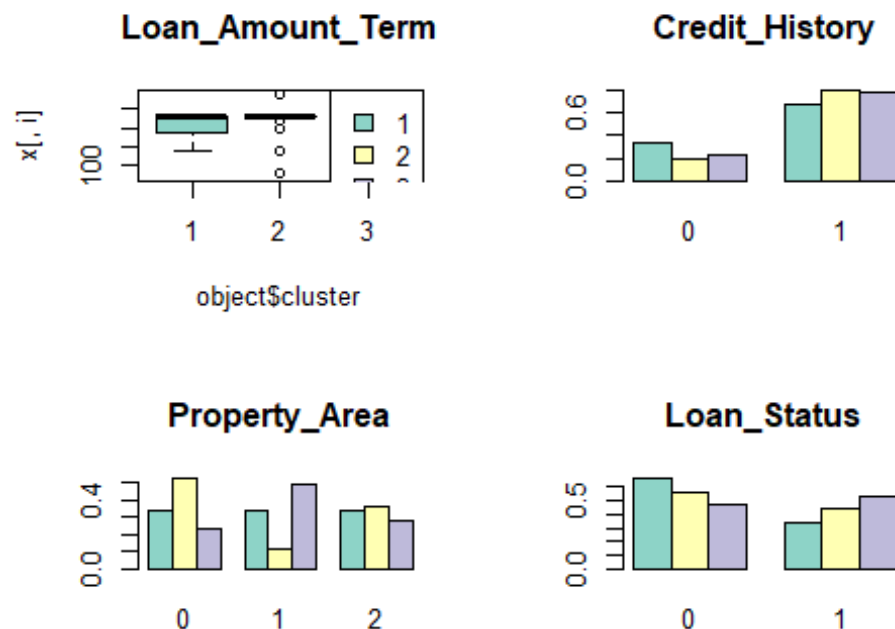


CoapplicantIncome



LoanAmount





In the plots, cluster 1 is represented in green, cluster 2 in yellow and cluster 3 in purple.

Cluster 1 (1 approved loan, 2 denied loan):

This cluster has a size of three points. It consists entirely of applicants that are male, married, graduates, and have no co-applicant income. The number of dependents vary and the property area in which they live is evenly distributed. Compared to the other clusters, they have the highest requested loan amount and the lowest loan amount term with a narrower range. In comparing this cluster's proportional distribution against the proportional distribution of the other two clusters, it has the most self-employed applicants and the most applicants with no credit history.

Many of the attributes are skewed due to this cluster containing a very small number of points. As a result, this cluster dominates many of the attributes. Considering this, for the next two clusters, we will compare them with each other and exclude cluster 1.

Cluster 2 (60 approved loan, 75 denied loan) and Cluster 3 (126 approved loan, 110 denied loan):

Cluster 2 has a size of 135 points and Cluster 3 has a size of 236 points. Proportionally, Cluster 2 contains more individuals that are female, unmarried, self-employed, and slightly more those with a credit history and that are graduates than

Cluster 3. On average, applicants in Cluster 2 have around 0 dependents and those in Cluster 3 have around 1. Compared to cluster 3, Cluster 2 has on average applicants with a higher income, lower co-applicant income, requested higher loan amount and within a wider range, and a slightly higher loan term with a narrower range. Cluster 2 had applicants that mostly live in urban areas and Cluster 3 has applicants that mostly live in semi-urban areas.

Gower Distance

The next method we will use is Gower distance. It can be used with class imbalance datasets and so we will use the processed loan data.

```
## Warning: package 'Rtsne' was built under R version 4.4.3
## Warning: package 'prcr' was built under R version 4.4.3
## 178503 dissimilarities, summarized :
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.0000717 0.2304500 0.3149900 0.3234600 0.4196800 0.8073300
## Metric :  mixed ;  Types = A, A, I, A, A, I, I, I, I, A, N
## Number of objects : 598
```

The plots in the visualization section show that the numeric variables Applicant Income, Dependents, coapplicant Income and Loan Amount are right skewed, so a log transformation will be applied to these variables when calculating the Gower distance. Additionally, we need to specify the variable types for the categorical variables. The binary variables are all asymmetric as was seen in the visualization section of this report, and so will be entered as asymmetric binary variables. As for property area, it will be considered as a factor. Since we want to exclude the target variable, Loan_Status, from the Gower distance calculation, we have removed it from the analysis.

We can display the most similar and dissimilar pairs of points to verify for any odd results.

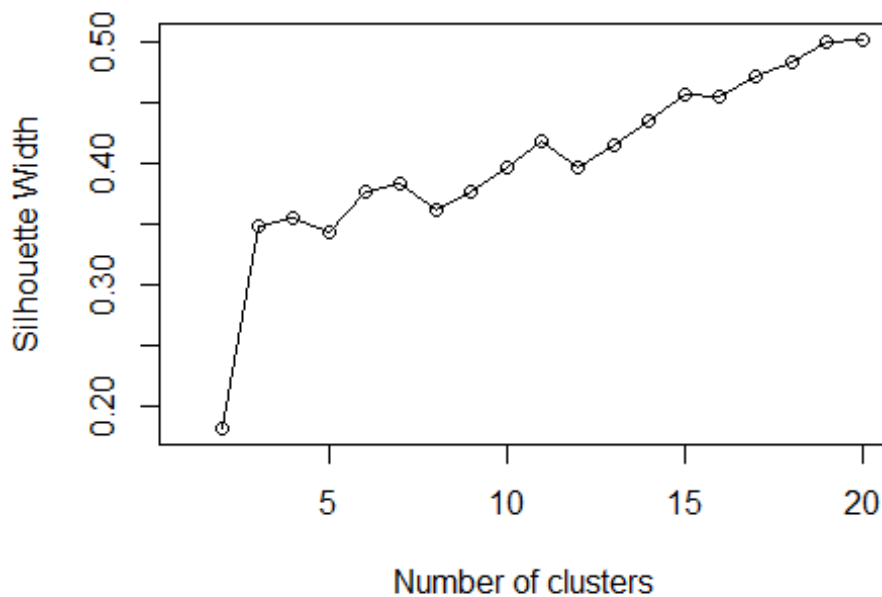
```
##      Gender Married Dependents Education Self_Employed
ApplicantIncome
## 225      1      1      1      1      0
2491
## 87      1      1      0      1      0
2500
##      CoapplicantIncome LoanAmount Loan_Amount_Term Credit_History
Property_Area
## 225      2054      104      360      1
1
## 87      2118      104      360      1
1
##      Loan_Status
```

```
## 225      1
## 87       1
```

The most similar pair of points have many equal attributes or attributes with values that are close in range. Their attributes are also those that are most common in the data. Both are applicants with an approved loan status.

```
##      Gender Married Dependents Education Self_Employed
ApplicantIncome
## 534      1         0           0         0           0
3358
## 54       0         1           1         1           1
11500
##      CoapplicantIncome LoanAmount Loan_Amount_Term Credit_History
Property_Area
## 534              0         80              36              1
1
## 54              0        286             360              0
0
##      Loan_Status
## 534              0
## 54              0
```

As for the dissimilar pair all their attributes are opposing and have a great difference between their values. However, both are applicants with a denied loan status.



We want to choose the number of cluster with the highest silhouette width. Using the elbow method, we will select $k = 3$.

Let's view the summary and the medoids:

```
## [[1]]
## Gender Married Dependents Education Self_Employed
ApplicantIncome
## 0: 33 0: 69 Min. :0.0000 0: 40 0:163 Min. :
416
## 1:165 1:129 1st Qu.:0.0000 1:158 1: 35 1st Qu.:
2611
## Median :0.0000 Median :
3445
## Mean :0.7677 Mean :
5159
## 3rd Qu.:1.0000 3rd Qu.:
5772
## Max. :3.0000 Max.
:63337
## CoapplicantIncome LoanAmount Loan_Amount_Term Credit_History
## Min. : 0 Min. : 9.00 Min. : 12.0 0: 31
## 1st Qu.: 0 1st Qu.: 93.25 1st Qu.:360.0 1:167
## Median : 1035 Median :120.00 Median :360.0
## Mean : 1738 Mean :139.48 Mean :333.8
## 3rd Qu.: 2160 3rd Qu.:160.00 3rd Qu.:360.0
## Max. :41667 Max. :650.00 Max. :480.0
## Property_Area Loan_Status cluster
## 0:198 0: 68 Min. :1
## 1: 0 1:130 1st Qu.:1
## 2: 0 Median :1
## Mean :1
## 3rd Qu.:1
## Max. :1
##
## [[2]]
## Gender Married Dependents Education Self_Employed
ApplicantIncome
## 0: 24 0: 63 Min. :0.0000 0: 47 0:140 Min. :
150
## 1:151 1:112 1st Qu.:0.0000 1:128 1: 35 1st Qu.:
2950
## Median :0.0000 Median :
3987
## Mean :0.7429 Mean :
5508
## 3rd Qu.:2.0000 3rd Qu.:
6022
## Max. :3.0000 Max.
:81000
```

```

## CoapplicantIncome    LoanAmount    Loan_Amount_Term Credit_History
## Min.      : 0        Min.      : 40.0    Min.      : 84.0    0: 27
## 1st Qu.: 0        1st Qu.:111.0    1st Qu.:360.0    1:148
## Median :1508      Median :135.0    Median :360.0
## Mean    :1631      Mean    :150.9    Mean    :345.1
## 3rd Qu.:2516      3rd Qu.:170.0    3rd Qu.:360.0
## Max.    :8333      Max.    :570.0    Max.    :480.0
## Property_Area Loan_Status    cluster
## 0: 0          0: 67        Min.    :2
## 1: 0          1:108       1st Qu.:2
## 2:175                Median :2
##                      Mean    :2
##                      3rd Qu.:2
##                      Max.    :2
##
## [[3]]
## Gender Married    Dependents    Education Self_Employed
ApplicantIncome
## 0: 54    0: 78    Min.    :0.0000    0: 46    0:185    Min.    :
210
## 1:171    1:147    1st Qu.:0.0000    1:179    1: 40    1st Qu.:
2927
##                      Median :0.0000    Median :
3858
##                      Mean    :0.7689    Mean    :
5242
##                      3rd Qu.:1.0000    3rd Qu.:
5266
##                      Max.    :3.0000    Max.
:39999
## CoapplicantIncome    LoanAmount    Loan_Amount_Term Credit_History
## Min.      : 0        Min.      : 25.0    Min.      : 36.0    0: 28
## 1st Qu.: 0        1st Qu.:100.0    1st Qu.:360.0    1:197
## Median : 997      Median :128.0    Median :360.0
## Mean    :1538      Mean    :145.2    Mean    :346.6
## 3rd Qu.:2253      3rd Qu.:164.0    3rd Qu.:360.0
## Max.    :20000     Max.    :600.0    Max.    :480.0
## Property_Area Loan_Status    cluster
## 0: 0          0: 52        Min.    :3
## 1:225          1:173       1st Qu.:3
## 2: 0                Median :3
##                      Mean    :3
##                      3rd Qu.:3
##                      Max.    :3

```

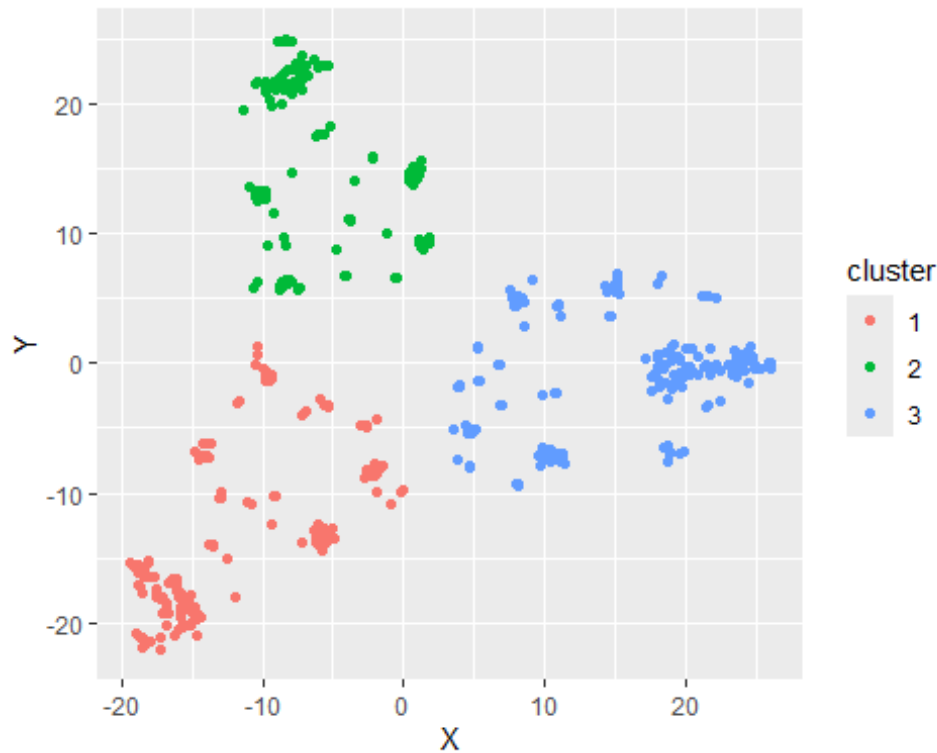
Based on the cluster summaries, the model did not produce strongly distinct groupings with respect to loan status. They have instead perfectly clustered by property area.

Examining the medoids, which are the most representative data points within each cluster, provides some insight. The medoids are as follows:

##	Gender	Married	Dependents	Education	Self_Employed
ApplicantIncome					
## 387	1	1	2	1	0
3100					
## 2	1	1	1	1	0
4583					
## 213	1	1	0	1	0
3727					
##	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area
## 387	1400	113	360	1	
0					
## 2	1508	128	360	1	
2					
## 213	1775	131	360	1	
1					
##	Loan_Status				
## 387	1				
## 2	0				
## 213	1				

Although many binary variables (Gender, Married, Education, Self-Employed, and Credit History) are asymmetric and do not vary across medoids, some distinctions emerge in the other features.

The most notable differences between the medoids lie in the number of dependents, the range of applicant and co-applicant incomes, and the property area. The medoid for Cluster 1 has 2 dependents, the lowest applicant and co-applicant incomes among the three, and an urban property area. The medoid for Cluster 3 has 0 dependents, the highest applicant and co-applicant incomes, and a semi-urban property area. Meanwhile, the medoid for Cluster 2 falls between the other two in terms of income, has 1 dependent, and is associated with a rural property area.



The clusters are very well-separated, with no overlapping points. However this clustering has reflected more upon the property area of the applicants rather than their loan status. Cluster 1 and 3 which represent the approved loans are further south on the graph than Cluster 2 which represents the denied loans.

Hence, the Gower distance clustering method did not effectively separate the clusters based on loan status but instead their property area. The medoids did not differ greatly but did identify two clusters that are primarily associated with approved loans and one for not approved.

Supervised Learning

The goal of supervised learning in this project is to develop predictive models that classify loan applications as approved or denied based on applicant attributes. We compare three algorithms, logistic regression, decision tree, and random forest using metrics such as accuracy, sensitivity, specificity, and area under the ROC curve. This comparison highlights trade-offs between predictive performance and interpretability to inform model selection for real-world deployment.

Logistic

We begin with a logistic regression model to estimate the probability of loan approval as a function of all predictor variables.

```
##
## Call:
## glm(formula = Loan_Status ~ ., family = binomial, data =
train.pcadata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.774541   0.387653  -4.578 4.70e-06 ***
## PC1          -0.245854   0.134887  -1.823 0.068355 .
## PC2           0.137422   0.141562   0.971 0.331669
## PC3           0.198253   0.141192   1.404 0.160278
## PC4           0.179926   0.148572   1.211 0.225884
## PCB1         -0.210327   0.038000  -5.535 3.11e-08 ***
## PCB2          0.119837   0.034495   3.474 0.000513 ***
## PCB3          0.006081   0.034155   0.178 0.858696
## PCB4          0.215232   0.041682   5.164 2.42e-07 ***
## Property_Area1 0.927579   0.333123   2.784 0.005361 **
## Property_Area2 -0.033277   0.343280  -0.097 0.922776
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 415.89  on 299  degrees of freedom
## Residual deviance: 327.49  on 289  degrees of freedom
## AIC: 349.49
##
## Number of Fisher Scoring iterations: 5
```

The intercept is highly significant, equal to the sample mean for the urban property area and the sample mean of the response when the PCs (numeric variables) are equal to 0.

Out of all the predictors, we have 4 that are significant; PCB1, PCB2, PCB4, and semi-urban property area. This model suggest that the PCs have very little statistical significance in relation to the response. Hence, PCA on numeric variables did not improve predictive ability.

PCB1: Dominated by Gender, Self_Employed, Education, and Credit_History
Estimate: -0.210327

Interpretation: A higher score on PCB1 (e.g., being male, self-employed, less educated, or lacking credit history) is associated with a decrease in the log-odds of loan approval.

PCB2: Primarily influenced by Marital Status and Credit_History Estimate: 0.119837

Interpretation: A higher PCB2 score (e.g., being unmarried with a good credit history) correlates with an increase in the log-odds of loan approval.

PCB4: Driven by Self_Employed and Credit_History Estimate: 0.215232

Interpretation: Individuals who are self-employed and have a credit history (higher PCB4 scores) are more likely to have their loans approved.

All three principal-component predictors are highly statistically significant in the loan-approval model at the $\alpha=0.001$ level. This robust evidence confirms that the underlying factors: gender, self-employment, education, marital status, and credit history which captured by these components exert a strong and reliable influence on the log-odds of loan approval.

```
## Loading required package: lattice

## Confusion Matrix and Statistics
##
##           Reference
## Prediction 0  1
##           0 24 18
##           1 13 64
##
##               Accuracy : 0.7395
##               95% CI : (0.6511, 0.8156)
##       No Information Rate : 0.6891
##       P-Value [Acc > NIR] : 0.1374
##
##               Kappa : 0.4138
##
##  Mcnemar's Test P-Value : 0.4725
##
##       Sensitivity : 0.6486
##       Specificity : 0.7805
##       Pos Pred Value : 0.5714
##       Neg Pred Value : 0.8312
##       Prevalence : 0.3109
##       Detection Rate : 0.2017
##       Detection Prevalence : 0.3529
##       Balanced Accuracy : 0.7146
##
##       'Positive' Class : 0
##
```

True Positives (TP) = 24

True Negatives (TN) = 64

False Positives (FP) = 18

False Negatives (FN) = 13

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##      cov, smooth, var

## Setting levels: control = 0, case = 1

## Setting direction: controls < cases

##              Value
## Accuracy  0.7394958
## Recall    0.6486486
## Precision 0.5714286
## F1        0.6075949
## AUC       0.7956493
```

Fair Recall (65%)

We're catching only about 65% of the actual approvals, whereas 35% of valid "approve" cases are slipping through as false negatives.

Less Than Fair Precision (57%)

When the model says "approve," it's correct roughly 57% of the time, meaning 43% of those flagged get rejected in reality. This result is not ideal as it is just above random.

Fair F1 (61%)

Precision and recall are slightly distanced, yielding an F1 of 61%. There is some bias toward one error type.

Moderate Accuracy (74%)

Good overall, but reflects the underlying class imbalance (around 74% approvals).

Good AUC (80%)

The model has good discriminative power, but there's room for improvement. Ideally an AUC of above 0.80 has stronger separation.

Tree Classification

Next, we train a decision tree classifier using the rpart package.

```
## Call:
## rpart(formula = Loan_Status ~ ., data = train.pcadata, method =
## "class",
##       minbucket = 10)
##       n= 300
```

```

##
##          CP nsplit rel error      xerror      xstd
## 1 0.32666667      0 1.0000000 1.1066667 0.05740564
## 2 0.06000000      1 0.6733333 0.6733333 0.05456766
## 3 0.03333333      2 0.6133333 0.7533333 0.05595104
## 4 0.02666667      6 0.4800000 0.7666667 0.05614136
## 5 0.02000000      7 0.4533333 0.7600000 0.05604760
## 6 0.01000000     10 0.3933333 0.7600000 0.05604760
##
## Variable importance
##          PCB4          PCB2          PC3          PC4
PC1
##          34          23          11          11
7
##          PCB1          PC2 Property_Area          PCB3
##          4          4          3          2
##
## Node number 1: 300 observations,      complexity param=0.3266667
##   predicted class=0   expected loss=0.5   P(node) =1
##   class counts:   150   150
##   probabilities: 0.500 0.500
##   left son=2 (53 obs) right son=3 (247 obs)
##   Primary splits:
##       PCB4          < -3.549587   to the left,   improve=27.511270,
(0 missing)
##       PCB2          < -4.137728   to the left,   improve=19.988930,
(0 missing)
##       PCB1          < -9.367836   to the right,  improve=12.311800,
(0 missing)
##       PC2           < -0.4297704   to the left,   improve= 4.652226,
(0 missing)
##       Property_Area splits as LRL, improve= 4.436610, (0 missing)
##   Surrogate splits:
##       PCB2 < -4.137728   to the left,   agree=0.89, adj=0.377, (0
split)
##
## Node number 2: 53 observations
##   predicted class=0   expected loss=0.03773585   P(node) =0.1766667
##   class counts:      51      2
##   probabilities: 0.962 0.038
##
## Node number 3: 247 observations,      complexity param=0.06
##   predicted class=1   expected loss=0.4008097   P(node) =0.8233333
##   class counts:      99     148
##   probabilities: 0.401 0.599
##   left son=6 (11 obs) right son=7 (236 obs)
##   Primary splits:
##       PCB2          < -5.127029   to the left,   improve=5.948613, (0
missing)
##       PC2           < -0.8511271   to the left,   improve=4.409474, (0

```

```

missing)
##      Property_Area splits as  LRL, improve=4.197571, (0 missing)
##      PCB1          < -9.367836  to the right, improve=4.193555, (0
missing)
##      PC4           < -0.2650201 to the left,  improve=3.418986, (0
missing)
##      Surrogate splits:
##      PCB4 < 7.906357    to the right, agree=0.972, adj=0.364, (0
split)
##
## Node number 6: 11 observations
##      predicted class=0 expected loss=0.09090909 P(node) =0.03666667
##      class counts:    10     1
##      probabilities: 0.909 0.091
##
## Node number 7: 236 observations,    complexity param=0.03333333
##      predicted class=1 expected loss=0.3771186 P(node) =0.7866667
##      class counts:    89    147
##      probabilities: 0.377 0.623
##      left son=14 (99 obs) right son=15 (137 obs)
##      Primary splits:
##      PC4           < -0.2650201 to the left,  improve=3.958482, (0
missing)
##      Property_Area splits as  LRL, improve=3.808084, (0 missing)
##      PC3           < -1.017619  to the left,  improve=3.734828, (0
missing)
##      PC2           < -0.8511271 to the left,  improve=3.254363, (0
missing)
##      PCB1          < -9.367836  to the right, improve=2.825670, (0
missing)
##      Surrogate splits:
##      PC3 < -0.5390118  to the left,  agree=0.691, adj=0.263, (0
split)
##      PC2 < 0.6158642   to the right, agree=0.619, adj=0.091, (0
split)
##      PCB1 < -5.971355  to the right, agree=0.614, adj=0.081, (0
split)
##      PCB2 < 2.046518   to the right, agree=0.614, adj=0.081, (0
split)
##      PC1 < 1.035549    to the right, agree=0.610, adj=0.071, (0
split)
##
## Node number 14: 99 observations,    complexity param=0.03333333
##      predicted class=1 expected loss=0.4848485 P(node) =0.33
##      class counts:    48     51
##      probabilities: 0.485 0.515
##      left son=28 (60 obs) right son=29 (39 obs)
##      Primary splits:
##      Property_Area splits as  LRL, improve=2.9545450, (0 missing)
##      PC3           < 1.202442   to the left,  improve=2.5564110,

```

```

(0 missing)
##      PC1          < -0.6822706  to the right, improve=1.5714290,
(0 missing)
##      PC4          < -1.101942   to the right, improve=0.8311688,
(0 missing)
##      PC2          < -0.07099707 to the left,  improve=0.8085123,
(0 missing)
## Surrogate splits:
##      PC1 < -0.8918947  to the right, agree=0.657, adj=0.128, (0
split)
##      PC2 < -1.631011   to the right, agree=0.636, adj=0.077, (0
split)
##      PCB4 < -0.5735653  to the right, agree=0.636, adj=0.077, (0
split)
##      PC3 < 2.085858     to the left,  agree=0.626, adj=0.051, (0
split)
##      PC4 < -2.191836   to the right, agree=0.626, adj=0.051, (0
split)
##
## Node number 15: 137 observations,    complexity param=0.03333333
## predicted class=1 expected loss=0.2992701 P(node) =0.4566667
## class counts:    41    96
## probabilities: 0.299 0.701
## left son=30 (100 obs) right son=31 (37 obs)
## Primary splits:
##      PC4 < -0.1700949  to the right, improve=3.704719, (0 missing)
##      PC3 < -0.7485482  to the left,  improve=3.550521, (0 missing)
##      PCB1 < -9.367836   to the right, improve=2.535242, (0 missing)
##      PC2 < -0.4297704   to the left,  improve=2.274305, (0 missing)
##      PC1 < -0.01452914  to the right, improve=1.833157, (0 missing)
## Surrogate splits:
##      PC3 < -0.4888611  to the right, agree=0.876, adj=0.541, (0
split)
##      PC1 < -0.5692941  to the right, agree=0.818, adj=0.324, (0
split)
##      PCB3 < 7.039228    to the left,  agree=0.752, adj=0.081, (0
split)
##      PCB4 < 7.457676    to the left,  agree=0.737, adj=0.027, (0
split)
##
## Node number 28: 60 observations,    complexity param=0.02
## predicted class=0 expected loss=0.4166667 P(node) =0.2
## class counts:    35    25
## probabilities: 0.583 0.417
## left son=56 (10 obs) right son=57 (50 obs)
## Primary splits:
##      PC1 < 1.079865     to the right, improve=2.406667, (0 missing)
##      PC3 < 1.125561     to the left,  improve=1.926667, (0 missing)
##      PCB2 < 5.447452    to the left,  improve=1.396489, (0 missing)
##      PC2 < -0.5051791   to the left,  improve=1.126667, (0 missing)

```

```

##      PCB1 < -7.075476   to the right, improve=0.952381, (0 missing)
##  Surrogate splits:
##      PC3 < -0.9845813   to the left,  agree=0.900, adj=0.4, (0
split)
##      PC2 < 0.9181052    to the right, agree=0.867, adj=0.2, (0
split)
##
## Node number 29: 39 observations,    complexity param=0.02666667
##  predicted class=1  expected loss=0.3333333  P(node) =0.13
##    class counts:    13    26
##    probabilities: 0.333 0.667
##  left son=58 (12 obs) right son=59 (27 obs)
##  Primary splits:
##      PCB2 < 5.673262    to the right, improve=3.8518520, (0
missing)
##      PCB4 < -0.479636   to the left,  improve=1.9500000, (0
missing)
##      PCB1 < -2.778995   to the right, improve=0.7471264, (0
missing)
##      PC1  < 0.244665    to the left,  improve=0.6190476, (0
missing)
##      PC3  < -0.5781564  to the left,  improve=0.5887681, (0
missing)
##  Surrogate splits:
##      PCB4 < -0.1104844  to the left,  agree=0.846, adj=0.500, (0
split)
##      PCB1 < -2.778995   to the right, agree=0.795, adj=0.333, (0
split)
##      PC4  < -0.3706062  to the right, agree=0.769, adj=0.250, (0
split)
##
## Node number 30: 100 observations,    complexity param=0.03333333
##  predicted class=1  expected loss=0.37  P(node) =0.3333333
##    class counts:    37    63
##    probabilities: 0.370 0.630
##  left son=60 (28 obs) right son=61 (72 obs)
##  Primary splits:
##      PC3  < -0.1304772  to the left,  improve=7.405714, (0 missing)
##      PC4  < 0.8184058   to the left,  improve=3.201169, (0 missing)
##      PCB2 < 3.199423    to the right, improve=2.874386, (0 missing)
##      PCB3 < -0.08623297 to the right, improve=2.859364, (0 missing)
##      PCB1 < -9.367836   to the right, improve=2.850110, (0 missing)
##  Surrogate splits:
##      PC2  < -0.1901307  to the left,  agree=0.80, adj=0.286, (0
split)
##      PC4  < -0.06646175 to the left,  agree=0.80, adj=0.286, (0
split)
##      PC1  < -0.9392967  to the left,  agree=0.77, adj=0.179, (0
split)
##      PCB3 < 6.016344    to the right, agree=0.75, adj=0.107, (0

```

```

split)
##
## Node number 31: 37 observations
##   predicted class=1   expected loss=0.1081081   P(node) =0.1233333
##   class counts:      4      33
##   probabilities: 0.108 0.892
##
## Node number 56: 10 observations
##   predicted class=0   expected loss=0.1   P(node) =0.03333333
##   class counts:      9      1
##   probabilities: 0.900 0.100
##
## Node number 57: 50 observations,   complexity param=0.02
##   predicted class=0   expected loss=0.48   P(node) =0.1666667
##   class counts:      26      24
##   probabilities: 0.520 0.480
##   left son=114 (36 obs) right son=115 (14 obs)
##   Primary splits:
##       PCB2 < 5.447452   to the left,   improve=2.134603, (0 missing)
##       PCB1 < -7.075476  to the right,  improve=1.940769, (0 missing)
##       PCB4 < 0.3381961  to the right,  improve=1.652699, (0 missing)
##       PC2  < -0.07047056 to the left,   improve=1.320294, (0 missing)
##       PCB3 < 4.454032   to the right,  improve=1.211748, (0 missing)
##   Surrogate splits:
##       PCB4 < -0.4601286 to the right,  agree=0.90, adj=0.643, (0
split)
##       PCB1 < -2.778995  to the left,   agree=0.78, adj=0.214, (0
split)
##       PC1  < -0.90284   to the right,  agree=0.74, adj=0.071, (0
split)
##
## Node number 58: 12 observations
##   predicted class=0   expected loss=0.3333333   P(node) =0.04
##   class counts:      8      4
##   probabilities: 0.667 0.333
##
## Node number 59: 27 observations
##   predicted class=1   expected loss=0.1851852   P(node) =0.09
##   class counts:      5      22
##   probabilities: 0.185 0.815
##
## Node number 60: 28 observations
##   predicted class=0   expected loss=0.3214286   P(node) =0.09333333
##   class counts:      19      9
##   probabilities: 0.679 0.321
##
## Node number 61: 72 observations
##   predicted class=1   expected loss=0.25   P(node) =0.24
##   class counts:      18      54
##   probabilities: 0.250 0.750

```

```

##
## Node number 114: 36 observations,    complexity param=0.02
##   predicted class=0   expected loss=0.3888889   P(node) =0.12
##   class counts:      22    14
##   probabilities: 0.611 0.389
##   left son=228 (21 obs) right son=229 (15 obs)
##   Primary splits:
##       PCB1 < -8.671656   to the right, improve=2.292063, (0 missing)
##       PC4  < -0.5027135  to the right, improve=2.000000, (0 missing)
##       PCB2 < 0.1602114   to the right, improve=1.396825, (0 missing)
##       PC1  < -0.2751827  to the left,  improve=1.272102, (0 missing)
##       PC3  < 0.6252984   to the left,  improve=1.234188, (0 missing)
##   Surrogate splits:
##       PCB4 < 0.3381961   to the right, agree=0.889, adj=0.733, (0
split)
##       PCB3 < -0.08623297 to the right, agree=0.833, adj=0.600, (0
split)
##       PCB2 < 0.1602114   to the right, agree=0.806, adj=0.533, (0
split)
##       PC1  < -0.0404203  to the left,  agree=0.778, adj=0.467, (0
split)
##       PC2  < -0.03548959 to the right, agree=0.778, adj=0.467, (0
split)
##
## Node number 115: 14 observations
##   predicted class=1   expected loss=0.2857143   P(node) =0.04666667
##   class counts:       4    10
##   probabilities: 0.286 0.714
##
## Node number 228: 21 observations
##   predicted class=0   expected loss=0.2380952   P(node) =0.07
##   class counts:      16    5
##   probabilities: 0.762 0.238
##
## Node number 229: 15 observations
##   predicted class=1   expected loss=0.4   P(node) =0.05
##   class counts:       6    9
##   probabilities: 0.400 0.600

```

The decision-tree model distilled the 300 loan records into a seven-split structure that is both interpretable and reasonably accurate:

Key drivers: Two transformed features—PCB4 and PCB2—account for almost 80% of the total split improvement. A very low PCB4 (< -3.55) alone flags a loan as almost certainly bad (node2, error $\approx 3\%$), while an extremely low PCB2 (< -5.13) performs the same gate-keeping role for the rest of the portfolio (node6).

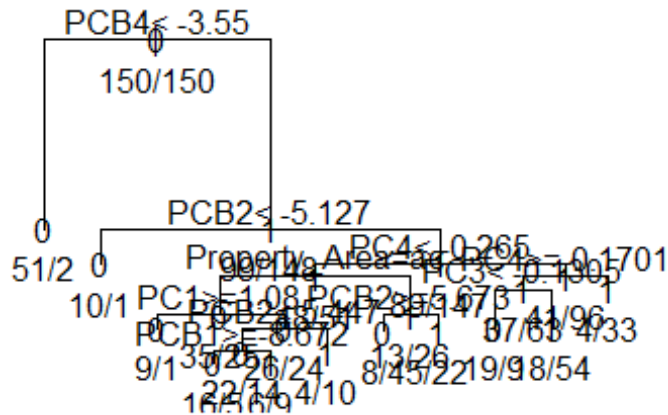
Refinements: Applicants who clear those two hurdles are sorted by PC4, Property_Area, and, to a lesser extent, PC3/PC1. These secondary splits raise the

posterior probability of a good loan (class1) from roughly 60% at node3 to nearly 90% in the “sweet-spot” leaf.

Model size vs reliability Cross-validation points to the subtree with CP=0.02 as the sweet spot: it trims three low-value splits without sacrificing predictive power, holding the 10-fold mis-classification rate at $\approx 33\%$. Keeping additional splits ($CP < 0.02$) yields no measurable gain.

Performance perspective: A 33% error rate is in line with the 25–40% leaf-level losses shown in the print-out, suggesting the tree generalises about as well as it fits. Compared with our earlier k-NN and logistic models, the tree trades a few points of accuracy for full transparency in how each decision is reached.

Classification Tree using rpart()



The pruned classification tree shows that loans with very negative PCB4 values (< -3.55), or if that hurdle is cleared—very negative PCB2 values (< -5.13), are overwhelmingly likely to default, whereas the remaining applicants are progressively sorted by property area and mid-range PC4/PC3 thresholds into branches whose default risk steadily declines, ending in a semi-urban subgroup with moderate component scores where roughly 9 out of 10 loans succeed, thereby cleanly segmenting the entire portfolio into high, medium, and low risk tiers with just seven easy to explain rules.

```
## 0 1
## 45 74
```

```
##  
## -1  0  1  
## 24 79 16
```

This output is showing, for each test case, the difference: predicted_class - actual_class -1 (count=24) Predicted 0 (deny) but actual1 (approve) => false negatives (we missed 24 real approvals).

0 (count=79) Predicted = actual => correct predictions (79 in total).

1 (count=16) Predicted 1 (approve) but actual 0 (deny) => false positives (we over-approved 16 denials).

```
## Confusion Matrix and Statistics  
##  
##           Reference  
## Prediction  0  1  
##           0 21 24  
##           1 16 58  
##  
##           Accuracy : 0.6639  
##           95% CI : (0.5715, 0.7478)  
##    No Information Rate : 0.6891  
##    P-Value [Acc > NIR] : 0.7579  
##  
##           Kappa : 0.2595  
##  
##    McNemar's Test P-Value : 0.2684  
##  
##           Sensitivity : 0.5676  
##           Specificity : 0.7073  
##           Pos Pred Value : 0.4667  
##           Neg Pred Value : 0.7838  
##           Prevalence : 0.3109  
##           Detection Rate : 0.1765  
##    Detection Prevalence : 0.3782  
##           Balanced Accuracy : 0.6374  
##  
##           'Positive' Class : 0  
##
```

Confusion Matrix

True negatives (correctly predicted approvals) = 58

True positives (correctly predicted denials) = 21

False positives (predicted deny but actually approve) = 24

False negatives (predicted approve but actually deny) = 16

With this test set the tree gets 66% overall accuracy, but because always guessing the majority class would already give 69%, the gain is statistically insignificant ($p \approx 0.76$) and agreement beyond chance remains only fair ($k \approx 0.26$); it manages to catch 57% of the truly bad loans (sensitivity) while correctly clearing 71% of the good ones (specificity), yet its low precision of 47% means that more than half the applicants it flags as “bad” are actually good, whereas its negative predictive value of 78% shows that borrowers it labels “good” are indeed safe nearly four times out of five, producing a balanced accuracy of 0.64 that signals moderate, but not compelling discriminatory power.

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
##           Value
## Accuracy  0.6638655
## Recall    0.5675676
## Precision  0.4666667
## F1        0.5121951
## AUC       0.6374423
```

Not beating the naive baseline: Because only 31% of applicants are actually bad, a trivial rule that rubber stamps everyone as good hits 69% accuracy, higher than the tree’s 66%. The McNemar and NIR p-values confirm the gap is not statistically significant.

Moderate recall, low precision: Recall 0.57 says the model intercepts only a bit more than half the truly risky loans. Precision 0.47 says that, of the applicants it does decline, more than half would have repaid—a costly loss of business.

Skew toward false alarms: The model is more willing to shout “risk” than to miss danger, but it still misses 16 bad loans and simultaneously blocks 24 good ones in this 119 loan test sample. That is a 1:1.5 trade-off between avoided defaults and lost revenue.

Discrimination is only “fair.” An AUC of 0.64 and balanced accuracy of 0.64 place the tree in the lower middle tier of credit risk models.

KNN

To complement our tree-based and regression approaches, we apply a k-nearest neighbors (kNN) classifier, which predicts loan approval by finding the k most similar past applications in feature space.

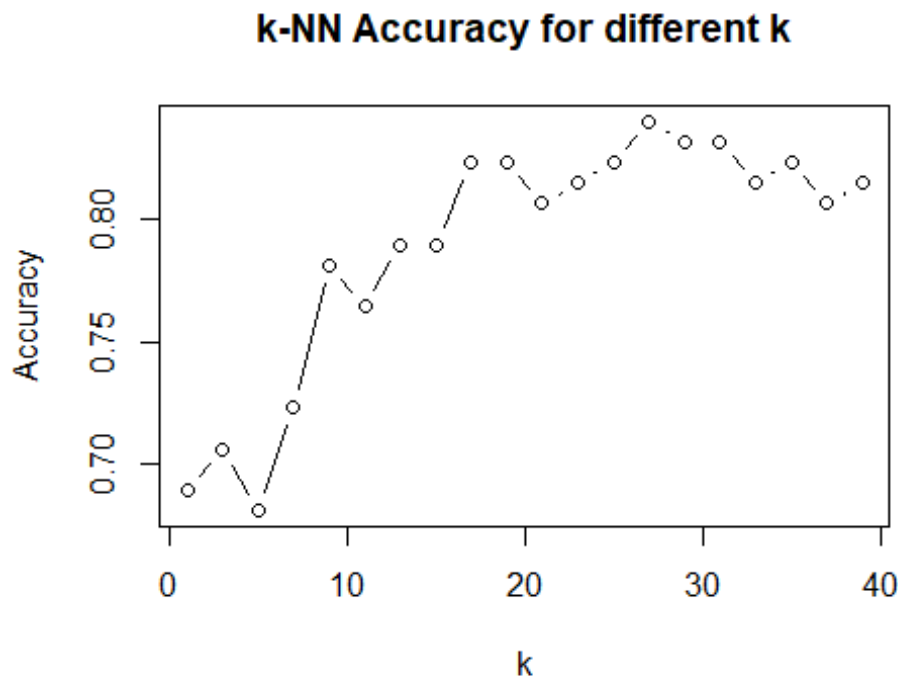
```
## [1] "k = 1 Accuracy = 0.689075630252101"
## [1] "k = 3 Accuracy = 0.705882352941176"
## [1] "k = 5 Accuracy = 0.680672268907563"
## [1] "k = 7 Accuracy = 0.722689075630252"
## [1] "k = 9 Accuracy = 0.781512605042017"
```

```

## [1] "k = 11 Accuracy = 0.764705882352941"
## [1] "k = 13 Accuracy = 0.789915966386555"
## [1] "k = 15 Accuracy = 0.789915966386555"
## [1] "k = 17 Accuracy = 0.823529411764706"
## [1] "k = 19 Accuracy = 0.823529411764706"
## [1] "k = 21 Accuracy = 0.80672268907563"
## [1] "k = 23 Accuracy = 0.815126050420168"
## [1] "k = 25 Accuracy = 0.823529411764706"
## [1] "k = 27 Accuracy = 0.840336134453782"
## [1] "k = 29 Accuracy = 0.831932773109244"
## [1] "k = 31 Accuracy = 0.831932773109244"
## [1] "k = 33 Accuracy = 0.815126050420168"
## [1] "k = 35 Accuracy = 0.823529411764706"
## [1] "k = 37 Accuracy = 0.80672268907563"
## [1] "k = 39 Accuracy = 0.815126050420168"

```

At $k=27$, the k -NN classifier reaches its peak accuracy of $\approx 84\%$, the highest of all tested values, placing it firmly in the optimal bias–variance regime.



$k \leq 7$ (high variance) – Accuracy hovers around 0.69-0.72 and even dips at $k=5$, showing that a very small neighborhood overfits to local noise.

$k=9-17$ (stabilizing) – As it let each query look at more neighbors, accuracy climbs sharply into the 0.78-0.82 band; the model is now smoothing out idiosyncratic points without yet blurring genuine class boundaries.

k=19-31(plateau) – The curve flattens at its high point: 0.82-0.84, peaking at k=27. Any value in this strip delivers virtually the same performance, so pick the lower end k=19 if we need faster predictions and the upper end k=27 if we need the single best score.

k≥33(rising bias) – Accuracy tilts down slightly toward 0.81 as the neighborhood grows too large and the classifier starts averaging across heterogeneous regions.

```
## [1] 1 1 0 1 1 1 1 1 1 1 1 1 1 1 0 1 1 0 0 1 1 1 1 1 1 1 1 0 1 1 1 1
1 1 1 1 1 1
## [38] 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 0 1 1 1 0 1
1 1 1 1 1 0
## [75] 1 1 0 1 1 0 0 1 0 1 1 1 0 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0 1 1 0
1 1 1 0 1 0
## [112] 1 1 1 1 1 0 1 1
## Levels: 0 1

## Confusion Matrix and Statistics
##
##           Reference
## Prediction 0  1
##           0 18  4
##           1 19 78
##
##               Accuracy : 0.8067
##               95% CI : (0.7242, 0.8734)
##           No Information Rate : 0.6891
##           P-Value [Acc > NIR] : 0.002775
##
##               Kappa : 0.4925
##
##  Mcnemar's Test P-Value : 0.003509
##
##           Sensitivity : 0.4865
##           Specificity : 0.9512
##           Pos Pred Value : 0.8182
##           Neg Pred Value : 0.8041
##           Prevalence : 0.3109
##           Detection Rate : 0.1513
##           Detection Prevalence : 0.1849
##           Balanced Accuracy : 0.7189
##
##           'Positive' Class : 0
##
```

True Positives (TP=18) – Loans the model flags as bad and that are indeed bad.

False Positives (FP=4) – Loans the model flags as bad but are actually good (costly lost business).

False Negatives (FN=19) – Bad loans the model mistakenly approves (credit-loss risk).

True Negatives (TN=78) – Good loans correctly approved.

This k-NN setup is exceptional at sparing good customers, only 4 false rejections—and reliable when it does decline someone. Its one shortcoming is that it still misses roughly half of the truly bad loans, so if default avoidance is paramount we can layer in a second check or adjust the decision threshold to boost recall. Overall, k-NN offers a strong trade off for customer-friendly credit decisions, with statistically significant lift over a naïve baseline.

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
##           Value
## Accuracy  0.8067227
## Recall    0.4864865
## Precision 0.8181818
## F1        0.6101695
## AUC       0.7188530
```

The k-NN model delivers a 0.81 overall accuracy and an AUC of 0.72, meaning it separates good versus bad loans substantially better than chance. Its precision of 0.82 tells us that when the model flags someone as “bad,” it is correct more than four-fifths of the time, so we avoid needless rejections. At the same time, its recall of 0.49 means it catches roughly half of all actual defaults, and the resulting F1 score of 0.61 reflects a solid but improvable balance between capturing risky loans and sparing good ones.

```
##           knn  logistic  tree
## Accuracy  0.8067227 0.7394958 0.6638655
## Recall    0.4864865 0.6486486 0.5675676
## Precision 0.8181818 0.5714286 0.4666667
## F1        0.6101695 0.6075949 0.5121951
## AUC       0.7188530 0.7956493 0.6374423
```

Across all five metrics the rpart tree clearly under-performs, so we can set it aside. The real contest is between logistic regression and k-NN:

k-NN

Accuracy=0.81 (best)

Precision=0.82 (best)

Recall=0.49 (lowest)

F1=0.61

AUC=0.72

k-NN excels at sparing good customers when it says “decline” it’s right 82% of the time—and delivers the highest overall hit rate, but it only catches half of the bad loans and its discrimination (AUC) is moderate.

Logistic regression

Accuracy=0.74

Precision=0.57

Recall=0.65 (best)

F1=0.61

AUC=0.80 (best)

Logistic regression offers the strongest ranking ability (AUC≈0.80) and the highest recall, so it intercepts nearly two thirds of defaults, even though it sacrifices some precision and overall accuracy.

Because both k-NN and logistic achieve essentially the same F1 (around 0.61), the choice hinges on business priorities:

If avoiding false declines is paramount (retaining good customers), k-NN is our top pick.

If catching as many defaults as possible is critical which means minimising loan losses, then logistic regression is preferable.

For a production system, we recommend logistic regression as the default, its superior AUC and recall give us the best risk separation, while tuning its probability threshold or combining it with a precision which is focused second check to control how many good customers we turn away.

Conclusion

In this project, we developed two unsupervised learning models; k-Prototypes and Gower Distance, and three supervised learning models; logistic regression, decision tree, and k-nearest neighbors (k-NN). These were done with the goal to predict loan approval.

First we preprocessed the data and applied two dimension reduction methods. The first was the standard PCA for the continuous variable. We selected the first 4 components that covered around 90% of the variability. The second method was logistic PCA for binary variables. We selected the first 4 components from this model that also covered around 90% of the variance in the data.

The unsupervised learning models were constructed using the processed data for Gower distance and the balanced processed data for k-Prototypes. The clusters created by both models did not succeed with separating very well by loan status. The k-Prototypes model had a number of both approved and rejected loan that were very close in range and created a cluster of only 3 points. The Gower Distance also did not succeed very well by clustering according to property area rather than loan status.

The supervised models were fitted using the balanced training set and the test set.

Data preprocessing & PCA

Raw borrower attributes: income, loan amount, term and dependents were transformed via principal components (PC1–PC4) to capture the main sources of variance.

Model training & evaluation

We fit each model on a 70/30 train/test split, measuring accuracy, recall (sensitivity), precision, F_1 , and AUC on the hold out set.

Model selection & business trade-off

The decision tree is easiest to explain but underperforms materially.

k-NN offers the highest accuracy and spares good customers which only has 4 false alarms, making it ideal where minimizing customer friction is critical.

Logistic regression strikes the best balance of recall and ranking powerkey when avoiding loan default carries the greatest cost.

Overall, our analysis suggests that logistic regression provides the best basis for producing loan approval scorecards, while k-NN can serve as an accuracy focused enhancement model in multistage decision-making processes.

data visualization: Selina Salameh

data reduction: Lening Gao

dimension reduction: Lening Gao (interpretations of components by both Lening Gao and Selina Salameh)

unsupervised learning: Selina Salameh

supervised learning: Xiwei Zhang